Special Theory of Relativity

Unit 1: Introduction to Special Relativity, Relativistic Kinematics and Dynamics
(with the Unit 3 topics on the Geometric Representation of Space-Time)

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Reference books

1. J. R. Taylor, *Classical Mechanics* (University Science Books, 2005), Ch. 15
2. J. V. Narlikar, *An Introduction to Relativity* (Cambridge Univ. Press, 2010), Chs. 1–3, 10
3. T. Takeuchi, *An Illustrated Guide to Relativity* (Cambridge Univ. Press, 2010)
4. B. W. Carroll & D. A. Ostlie, *An Introduction to Modern Astrophysics*, 2nd ed., Chs. 4, 17

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Part A Why Relativity Had to Be Invented

The goal of Part A.
By 1900 physics rested on two magnificent pillars: **Newton's mechanics** (1687) and **Maxwell's electromagnetism** (1865). Each was spectacularly successful on its own, but they disagreed about one simple question: *if I move, does light move past me more slowly?* Part A follows the detective story step by step – what each pillar says, where they clash, which experiment was designed to settle the clash, why it was performed, what it found, how Lorentz tried to rescue the old picture, and how Einstein finally cut the knot in 1905. Every later formula of relativity is an answer to a question raised here.

1 Frames of reference, events and the "relativity of measurements"

Almost every measurement in physics is made *relative* to something. The position $\mathbf{r} = (x, y, z)$ of a particle means its coordinates relative to a chosen origin and axes; the time $t = 5\,\mathrm{s}$ means $5\,\mathrm{s}$ after some chosen zero of time; the kinetic energy of a car is very different measured from the road and measured from inside the car. Taylor calls this the **relativity of measurements**, and the *theory of relativity* is simply the careful study of how measurements made by different observers are related (Taylor §15.1).

- An **event** is something that happens at a definite place and a definite instant – a firecracker exploding, a light flash being emitted, a muon decaying. It is labelled by four numbers (x, y, z, t).
- A **frame of reference** is a set of axes $Oxyz$ together with a set of clocks that assigns these four numbers to every event.
- An **inertial frame** is one in which Newton's first law holds: a body on which no net force acts moves with constant velocity. A frame fixed to the Earth is (to a good approximation) inertial; a frame fixed to a spinning turntable or an accelerating rocket is not.

Einstein's relativity is really two theories. **Special relativity** (1905) is "special" because it deals only with *inertial* (unaccelerated) frames. **General relativity** (1915) includes accelerated frames and turns out to be the relativistic theory of gravity (we meet it briefly in Section 29).

2 The Galilean transformation and Newtonian relativity

2.1 The standard configuration

Consider two frames: S fixed to the ground, and S' fixed to a railway carriage moving with constant velocity V along the common x-axis (Fig. 1). The axes are parallel and the origins coincide at $t = t' = 0$. This arrangement is called the **standard configuration**.

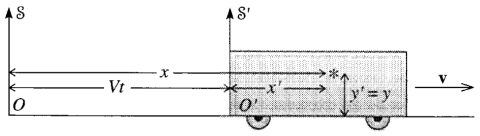


Figure 1: Frame S is fixed to the ground, frame S' to a railway car moving with constant velocity V along x. The star marks an event (a small explosion). From the figure, $x' = x - Vt$.
Source: J. R. Taylor, Classical Mechanics*, Fig. 15.1*

It was a basic assumption of Newtonian physics that there is a *single universal time* – clocks in all frames, once synchronised, agree forever. Reading off Fig. 1 for an event with coordinates (x, y, z, t) in S and (x', y', z', t') in S':

Galilean transformation (standard configuration).

$$x' = x - Vt, \qquad y' = y, \qquad z' = z, \qquad t' = t.$$

Inverse: $x = x' + Vt'$, $y = y'$, $z = z'$, $t = t'$. For a general direction of $\mathbf{V}$: $\mathbf{r}' = \mathbf{r} - \mathbf{V}t$, $t' = t$.

2.2 Consequences: what is the same and what is different

Differentiating with respect to time (remember $t' = t$ and V is constant):

$$\mathbf{v}' = \mathbf{v} - \mathbf{V} \qquad \text{(classical velocity-addition formula)}, \qquad \mathbf{a}' = \mathbf{a}.$$

- **Invariant** (same in both frames): time intervals Δt, lengths $\Delta x = x_2 - x_1$ measured at the same time, acceleration $\mathbf{a}$, mass m, force $\mathbf{F}$.
- **Not invariant**: position, velocity, momentum and kinetic energy. (The *conservation* laws still hold in every frame, but the conserved numbers differ.)

Derivation [5 marks]: Galilean invariance of Newton's second law
To show: if $\mathbf{F} = m\mathbf{a}$ holds in S, then $\mathbf{F}' = m'\mathbf{a}'$ holds in S'.

(i) From $\mathbf{r}' = \mathbf{r} - \mathbf{V}t$: $\mathbf{v}' = \mathrm{d}\mathbf{r}'/\mathrm{d}t' = \mathrm{d}\mathbf{r}/\mathrm{d}t - \mathbf{V} = \mathbf{v} - \mathbf{V}$ (since $t' = t$).

(ii) Differentiating again with $\mathbf{V}$ constant: $\mathbf{a}' = \mathrm{d}\mathbf{v}'/\mathrm{d}t = \mathbf{a}$.

(iii) Mass is found experimentally to be the same in all frames: $m' = m$.

(iv) Forces depend only on relative separations and relative velocities of bodies (springs, gravity, contact forces), which are unchanged by the transformation: $\mathbf{F}' = \mathbf{F}$.

(v) Hence $\mathbf{F}' = \mathbf{F} = m\mathbf{a} = m'\mathbf{a}'$. Newton's second law has the *same form* in S'. (The first and third laws follow similarly.) ■

Newtonian (Galilean) principle of relativity.
The laws of *mechanics* are the same in all inertial frames. Consequently *no mechanical experiment* performed inside a closed laboratory can tell whether the laboratory is at rest or moving uniformly. There is no "absolute rest" as far as mechanics is concerned.

Galileo's ship (1632).

In his *Dialogue Concerning the Two Chief World Systems*, Galileo asked his reader to shut himself below deck in a large ship with flies, fish in a bowl and dripping water. Whether the ship is at anchor or sailing smoothly, "you will discover not the least change in all the effects named". He used this to answer critics who said a moving Earth would leave falling stones behind – the first clear statement of the principle of relativity.

Solved Example 1: Galilean velocity addition and momentum conservation

A train moves at $V = 20\,\mathrm{m\,s^{-1}}$. A ball of mass $0.5\,\mathrm{kg}$ rolls forward on the floor at $v' = 5\,\mathrm{m\,s^{-1}}$ relative to the train and collides elastically head-on with an identical ball at rest in the train.

(a) Ground velocity of the moving ball: $v = v' + V = 25\,\mathrm{m\,s^{-1}}$.

(b) Train frame: before, $p' = 0.5(5) + 0 = 2.5\,\mathrm{kg\,m\,s^{-1}}$; after the elastic collision of equal masses the velocities exchange, $p' = 0 + 0.5(5) = 2.5$. Conserved.

Ground frame: before, $p = 0.5(25) + 0.5(20) = 22.5$; after, $0.5(20) + 0.5(25) = 22.5$. Conserved.

The *values* of momentum differ in the two frames, but the *law* (conservation) holds in both – exactly what Newtonian relativity promises.

3 Incompatibility of electromagnetism with Newtonian relativity

3.1 Maxwell's equations contain a speed

In 1865 Maxwell combined the laws of Coulomb, Ampère and Faraday (plus his displacement current) into four equations. In empty space they predict waves travelling with speed

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3.00 \times 10^8\,\mathrm{m\,s^{-1}},$$

which is the measured speed of light: light is an electromagnetic wave. But notice: ε_0 and μ_0 are constants measured in the laboratory, and nothing in the equations says "relative to what". If Maxwell's equations hold in frame S, light moves at c in all directions *in* S.

3.2 Light speed under a Galilean transformation

Now look at the same light from the carriage S' (Fig. 2). By the classical velocity-addition rule $v' = v - V$, a beam travelling forward has speed $c - V$ in S' and a beam travelling backward has speed $c + V$. So if the Galilean transformation is correct, Maxwell's equations can be true in *only one* inertial frame – the frame in which light has the same speed c in every direction.

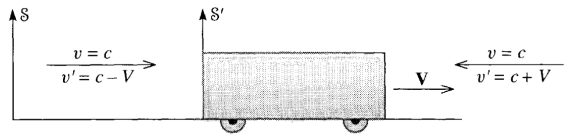


Figure 2: Two light beams approach the moving car from opposite directions. If light has speed c in S, the Galilean rule gives $c - V$ and $c + V$ in S': Maxwell's equations could then hold only in one special frame.

Source: J. R. Taylor, Classical Mechanics, *Fig. 15.2*

3.3 The wave equation changes its form

A cleaner way to see the clash (Narlikar §1.3) is to transform the wave equation itself. Every component ψ of $\mathbf{E}$ or $\mathbf{B}$ in vacuum obeys

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}-\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2}=0. \tag{1}$$

Derivation [5 marks]: The wave equation is not Galilean-invariant

Under $x'=x-Vt,\ y'=y,\ z'=z,\ t'=t$, the chain rule gives (since $\partial x'/\partial t=-V$, $\partial t'/\partial t=1$, $\partial x'/\partial x=1$, $\partial t'/\partial x=0$):

$$\frac{\partial}{\partial x}=\frac{\partial}{\partial x'},\qquad \frac{\partial}{\partial t}=\frac{\partial}{\partial t'}-V\frac{\partial}{\partial x'}.$$

Therefore

$$\frac{\partial^2}{\partial t^2}=\frac{\partial^2}{\partial t'^2}-2V\frac{\partial^2}{\partial x'\partial t'}+V^2\frac{\partial^2}{\partial x'^2},$$

and Eq. (1) becomes

$$\Big(1-\frac{V^2}{c^2}\Big)\frac{\partial^2\psi}{\partial x'^2}+\frac{\partial^2\psi}{\partial y'^2}+\frac{\partial^2\psi}{\partial z'^2}+\frac{2V}{c^2}\frac{\partial^2\psi}{\partial x'\partial t'}-\frac{1}{c^2}\frac{\partial^2\psi}{\partial t'^2}=0.$$

The extra terms (in V/c^2 and V^2/c^2) mean the equation *does not keep its form*. An observer in S' would find light travelling at different speeds in different directions, and could thereby detect his own "absolute" velocity V. ■

3.4 Two more symptoms

1. **Moving charges.** Two equal charges move side by side with the same velocity. In their own rest frame they only repel electrically (Coulomb). In the ground frame they also constitute two parallel currents, which attract magnetically, so the net repulsion is *smaller*. But Newtonian relativity insists that force is frame-independent ($\mathbf{F}'=\mathbf{F}$). Contradiction! (Unit 2 shows how relativity resolves it.)

2. **Magnet and conductor.** Moving a magnet towards a stationary loop, or moving the loop towards a stationary magnet, induces the same current; yet the 19th-century explanation was completely different in the two cases (an induced electric field in one, a magnetic force $q\mathbf{v}\times\mathbf{B}$ in the other). Einstein opened his 1905 paper with exactly this asymmetry.

Three possible ways out (as seen around 1900).

(a) The principle of relativity holds for mechanics, but electromagnetism has a **preferred frame** (the ether frame) – Galilean transformation correct, Maxwell valid only in that frame.

(b) A principle of relativity holds for *all* physics, but **Maxwell's equations are wrong** and must be modified to be Galilean-invariant.

(c) A principle of relativity holds for *all* physics, **Maxwell is right**, and it is **Newtonian mechanics and the Galilean transformation** that must be changed.

Option (b) failed – every attempt to modify Maxwell's equations predicted effects never observed. Option (a) was tested by Michelson and Morley. Option (c) is Einstein's special relativity.

4 The ether hypothesis

By the mid-19th century the wave nature of light was established by interference, diffraction and polarisation. Every wave then known needed a medium: water waves travel in water, sound in air, elastic waves in solids. So physicists assumed that light waves travel in an invisible medium filling all space – the **luminiferous ether** (from the Greek word for the upper, shining air).

The ether had to have strange properties:

- it fills even "empty" space, since starlight reaches us;
- it is extremely *rigid* (light is a transverse wave of enormous speed, and speed of transverse waves $\sim \sqrt{\text{rigidity}/\text{density}}$);
- yet it offers *no resistance* at all to planets ploughing through it;
- Maxwell's equations hold exactly in the frame where the ether is at rest – the **ether frame**, a candidate for Newton's "absolute space".

The Earth orbits the Sun at $v \approx 3 \times 10^4\,\mathrm{m\,s^{-1}}$, in a direction that changes through the year. It therefore cannot be at rest in the ether at all times; there should be an **ether wind** of about $30\,\mathrm{km\,s^{-1}}$ blowing past the Earth, and light should travel at slightly different speeds $(c \pm v)$ in different directions. Since $v/c \approx 10^{-4}$, the effect is tiny – but measurable, if one is clever.

4.1 Stellar aberration: the ether is not dragged along

In 1727 James Bradley found that every star traces a small ellipse in the sky over a year, of angular radius $\alpha \approx 20.5''$. The explanation is like rain falling vertically on a moving person: to catch the light in a telescope moving with the Earth at speed v, the telescope must be tilted forward by

$$\tan\alpha = \frac{v}{c} \approx \frac{3\times10^4}{3\times10^8} = 10^{-4}\,\mathrm{rad} = 20.6''.$$

Aberration shows that light from stars arrives in straight lines unaffected by the Earth's motion – the ether is *not* dragged along by the Earth near its surface (an "ether drag" would bend the light). Keep this in mind: it kills one escape route later.

Fizeau's water experiment (1851).
Fizeau sent light through water flowing at speed u. The light was speeded up, but only by the fraction $(1 - 1/n^2)u$ (Fresnel's "partial drag coefficient"), not by the full u. This odd result puzzled physicists for 50 years. In Section 16 you will see that it is exactly what the relativistic velocity-addition formula predicts.

5 The Michelson–Morley experiment (1887)

5.1 Why it was performed

The aim was to detect the Earth's motion through the ether – to *measure the ether wind*. First-order effects (of size v/c) cancel in most experiments where light goes out and back, so one must detect a second-order effect of size $v^2/c^2 \approx 10^{-8}$. Michelson invented an interferometer sensitive enough to do this (1881); with Edward Morley he repeated it with much greater precision in Cleveland in 1887.

5.2 The idea: a boat race on a river

Narlikar explains the experiment with a boatman who rows at speed c (relative to the water) on a river flowing at speed v (Fig. 3). Rowing a distance L downstream and back takes longer than rowing the same distance across the river and back – even though both are "round trips".

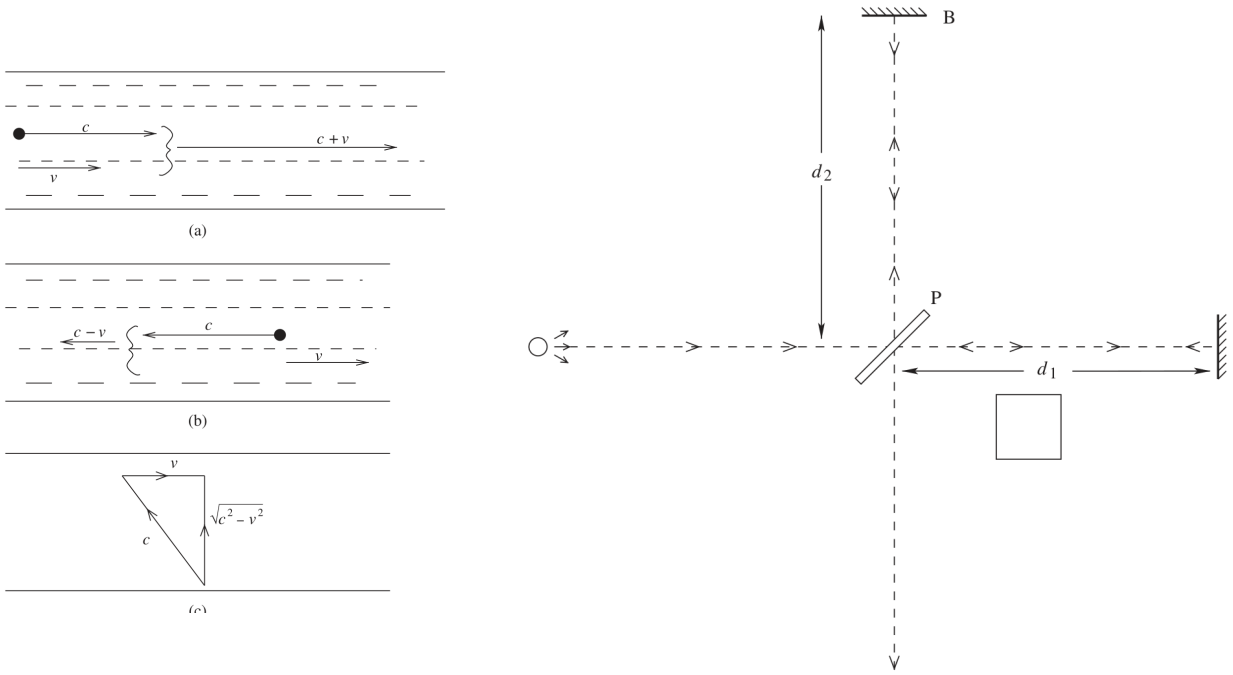


Figure 3: **Left:** a boat rowed with speed c on a river flowing with speed v: (a) downstream $c+v$, (b) upstream $c-v$, (c) across the river, net speed $\sqrt{c^2-v^2}$. **Right:** schematic of Michelson's interferometer: source S, half-silvered plate P, mirrors A and B at distances d_1 and d_2, and the observing telescope.

Source: J. V. Narlikar, An Introduction to Relativity*, Figs. 1.1 and 1.2*

5.3 The apparatus

Light from a source is split by a half-silvered plate P into two beams travelling along perpendicular arms of (nearly) equal length L. After reflection from mirrors at the ends, the beams recombine and produce interference fringes. If the travel times along the two arms differ by Δt, the fringe pattern is shifted by $c\Delta t/\lambda$ fringes. The whole apparatus rested on a massive sandstone slab floating in a trough of mercury, so it could be rotated smoothly through 90° while the observer watched the fringes. Multiple reflections made the effective arm length $L \approx 11\,\mathrm{m}$.

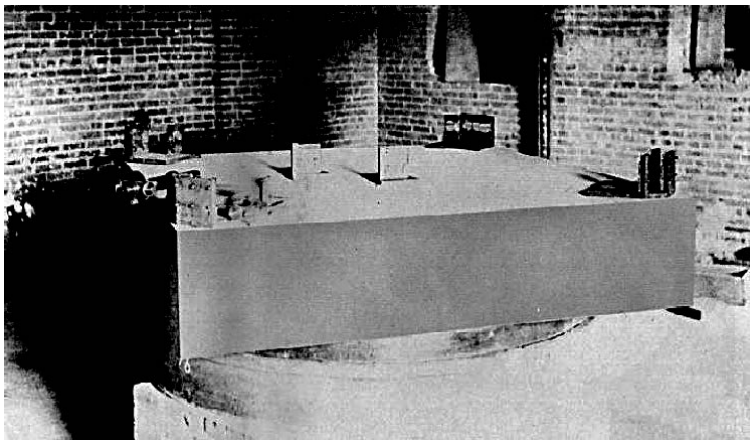


Figure 4: The 1887 Michelson–Morley interferometer: optics mounted on a stone slab floating on mercury; mirrors at the corners fold each beam back and forth to give an effective path of about 11 m.

Source: Wikimedia Commons, "Michelson morley experiment 1887.jpg" (public domain; from Michelson & Morley, Am. J. Sci. *34, 333, 1887)*

Derivation [5 marks]: Expected fringe shift in the Michelson–Morley experiment

Let the ether wind blow with speed v along arm 1 (length L); write $\beta = v/c$.

Arm parallel to the wind. Out: speed $c - v$; back: speed $c + v$.

$$t_{\parallel} = \frac{L}{c-v} + \frac{L}{c+v} = \frac{2Lc}{c^2 - v^2} = \frac{2L}{c}\,\frac{1}{1-\beta^2} \approx \frac{2L}{c}\big(1+\beta^2\big).$$

Arm perpendicular to the wind. Like the boat crossing the river, the light must be aimed slightly upstream, so its speed across is $\sqrt{c^2 - v^2}$ both ways:

$$t_{\perp} = \frac{2L}{\sqrt{c^2 - v^2}} = \frac{2L}{c}\,\frac{1}{\sqrt{1-\beta^2}} \approx \frac{2L}{c}\Big(1 + \tfrac{1}{2}\beta^2\Big).$$

Difference: $\Delta t = t_{\parallel} - t_{\perp} \approx \dfrac{L}{c}\,\beta^2 = \dfrac{Lv^2}{c^3}.$

Rotation by 90° interchanges the roles of the arms, so the time difference changes from $+\Delta t$ to $-\Delta t$: a total change $2\Delta t$. The number of fringes that should move past the cross-hair is

$$\boxed{\Delta N = \frac{c\,(2\Delta t)}{\lambda} = \frac{2L}{\lambda}\,\frac{v^2}{c^2}} \qquad \blacksquare$$

Solved Example 2: How big was the expected effect?

Take $L = 11\,\text{m}$, $\lambda = 5.9 \times 10^{-7}\,\text{m}$ (sodium light) and $v = 3 \times 10^{4}\,\text{m s}^{-1}$ ($\beta^2 = 10^{-8}$).

$$\Delta N = \frac{2(11)}{5.9 \times 10^{-7}} \times 10^{-8} = 0.37 \text{ fringe}.$$

The apparatus could detect a shift of about 0.01 fringe – some 40 times smaller than predicted. **Observed shift: less than 0.01 fringe, i.e. essentially zero.** The experiment was repeated at different times of day and in different seasons (six months apart the Earth's orbital velocity is reversed) with the same **null result**.

The null result.
There is no detectable ether wind. Light travels with the same speed in all directions in the Earth's frame, even though the Earth's velocity changes through the year. Equivalently: there is no experimental evidence for a unique "ether frame".

6 Attempts to save the ether

1. **Ether drag.** Perhaps the Earth drags the nearby ether along, as it drags its atmosphere, so there is no wind at the surface. *Killed by stellar aberration*: a dragged ether would bend starlight and destroy the aberration effect, which is observed.

2. **Emission (ballistic) theory** (Ritz, 1908). Perhaps light moves at c relative to its *source*, like a bullet, so on Earth it always has speed c. *Killed by binary stars* (de Sitter, 1913): light from the approaching star of a binary would overtake light from the receding star and the orbit would look scrambled; it does not. Modern tests with γ-rays from fast-moving π^0 mesons ($v = 0.99975c$, CERN 1964) found the photons still travel at c.

3. **Length contraction** (FitzGerald 1889, Lorentz 1892). Perhaps motion through the ether physically squeezes the arm along the wind by exactly the right factor. This is the idea that led directly to relativity – next section.

7 The contribution of H. A. Lorentz

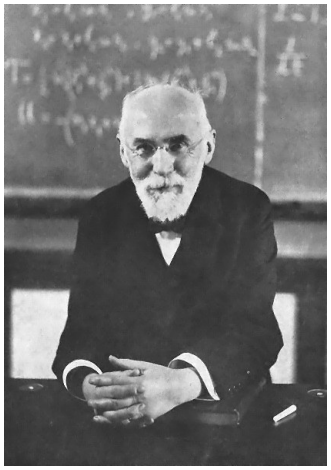


Figure 5: Hendrik Antoon Lorentz (1853–1928), Leiden. Nobel Prize 1902 (with Zeeman). He gave us the electron theory of matter, the force law $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$, the contraction hypothesis, "local time" and, by 1904, the full transformation of space and time that now bears his name. *Source: Wikimedia Commons, "Hendrik Antoon Lorentz.jpg" (public domain)*

Lorentz was the first physicist to realise that *to keep Maxwell's equations valid for a moving observer, both the space and the time coordinates must be transformed.* His work developed in stages:

1. **The electron theory (1878–1892).** Matter is made of charged particles ("electrons") moving in a stationary ether; all optical and electrical properties of matter come from them. In this picture the forces holding a solid together are electromagnetic.

2. **The contraction hypothesis (1892).** Independently of G. F. FitzGerald (1889), Lorentz proposed that a body moving with speed v through the ether is shortened along its motion by the factor $\sqrt{1 - v^2/c^2}$. Since intermolecular forces are electromagnetic and electromagnetic fields of moving charges are altered by motion, he argued that this contraction is a real *dynamical* effect of the ether wind on the molecules.

3. **"Local time" (1895).** To show that optical experiments on the moving Earth cannot detect the ether to first order in v/c, Lorentz introduced an auxiliary time variable

$$t_{\text{local}} = t - \frac{v\,x}{c^2},$$

which makes Maxwell's equations look the same in the moving frame to first order in v/c (his "theorem of corresponding states"). He regarded it as a mathematical aid, *not* as the time shown by real clocks.

4. **The full transformation (1899, 1904).** In his 1904 paper *Electromagnetic phenomena in a system moving with any velocity smaller than that of light*, Lorentz wrote the exact transformation that leaves Maxwell's equations unchanged:

$$x' = \gamma(x - vt), \quad y' = y, \quad z' = z, \quad t' = \gamma\Big(t - \frac{vx}{c^2}\Big), \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

(J. Larmor had obtained essentially the same equations in 1897–1900 and noticed that moving electron orbits would run slow – an early glimpse of time dilation.)

5. **Poincaré (1905)** named these the **Lorentz transformations**, showed that they form a group, corrected Lorentz's transformation of charge density and velocity, and stated a "principle of relativity" for all physics.

Derivation [5 marks]: Lorentz–FitzGerald contraction explains the null result

Suppose the arm moving *along* the ether wind is contracted from L to $L_c = L\sqrt{1-\beta^2}$, while the perpendicular arm is unchanged. Then

$$t_{\parallel} = \frac{2L_c}{c}\,\frac{1}{1-\beta^2} = \frac{2L\sqrt{1-\beta^2}}{c\,(1-\beta^2)} = \frac{2L}{c}\,\frac{1}{\sqrt{1-\beta^2}} = t_{\perp}.$$

The two times are *exactly* equal for every speed v and every orientation, so no fringe shift is expected on rotation – in agreement with Michelson and Morley. ■

Limitation: Kennedy and Thorndike (1932) used an interferometer with *unequal* arms. Then contraction alone is not enough – one also needs clocks (the light's frequency) to slow down by the same factor. They again found a null result: contraction *and* time dilation are both needed.

Lorentz vs. Einstein – same equations, different physics.

Lorentz (1904)	**Einstein (1905)**
The ether exists and defines absolute rest.	No ether is needed; no frame is special.
Contraction is a real squeezing of molecules by motion through the ether.	Contraction is a property of space and time measurements; it is mutual and symmetric.
t' is a fictitious "local time"; true time is absolute.	t' is *the* time read by real clocks in S'; simultaneity is relative.
Transformation derived by demanding Maxwell's equations keep their form.	Transformation derived from two simple postulates about *all* of physics.

Lorentz built the mathematical machinery; Einstein understood what it meant. That is why the transformation is often called the *Lorentz–Einstein* transformation (Taylor §15.6).

8 Einstein's postulates of special relativity

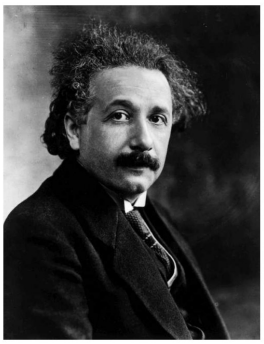


Figure 6: Albert Einstein (1879–1955). In 1905, while working at the patent office in Bern, he published *On the Electrodynamics of Moving Bodies*. Its starting point was not the Michelson–Morley result but his conviction that Maxwell's equations must hold in every inertial frame (Taylor, footnote to §15.3; Narlikar §1.2).
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 4, Fig. 3*

Following Taylor (§15.3), first agree what an inertial frame is: *an inertial frame is any frame in which all the laws of physics hold in their usual form.*

The two postulates.

Postulate 1 (Principle of relativity). If S is an inertial frame and a second frame S' moves with constant velocity relative to S, then S' is also an inertial frame. Equivalently: the laws of *all* physics – mechanics, electromagnetism, optics, … – are the same in all inertial frames. There is no absolute motion and no preferred frame.

Postulate 2 (Constancy of the speed of light). The speed of light in vacuum has the same value c in every direction in all inertial frames, independent of the motion of the source or the observer.

- Postulate 2 is the Michelson–Morley result raised to a law of nature (and it also follows from Postulate 1 + Maxwell's equations).
- Together they force us to give up the Galilean transformation and Newton's universal time. Every surprising effect that follows (relativity of simultaneity, time dilation, length contraction) is a *logical consequence* of these two statements.
- All the new effects depend on V^2/c^2 and become noticeable only when V is comparable with c. At everyday speeds Newton is an excellent approximation – which is why nobody noticed earlier.

Timeline: from Newton to Einstein.

1632	Galileo's ship: principle of relativity for mechanics.
1687	Newton's *Principia*: absolute space and absolute time.
1727	Bradley discovers stellar aberration ($20.5''$).
1851	Fizeau measures light speed in moving water (partial drag).
1865	Maxwell's equations; light is an electromagnetic wave with speed $1/\sqrt{\mu_0\varepsilon_0}$.
1881, 1887	Michelson, then Michelson & Morley: no ether wind.
1889, 1892	FitzGerald and Lorentz: contraction hypothesis.
1895	Lorentz: local time $t - vx/c^2$; Lorentz force law.
1897–1900	Larmor writes the transformation; notes slowing of moving electron orbits.
1904	Lorentz: full transformations leaving Maxwell's equations invariant.
1905	Poincaré names the Lorentz group; Einstein's special relativity and $E = mc^2$.
1908	Minkowski: "space by itself and time by itself are doomed to fade into mere shadows" – spacetime.
1932	Kennedy–Thorndike: unequal-arm interferometer, null result.
1938	Ives–Stilwell: transverse Doppler effect (time dilation of moving atoms).
1941	Rossi–Hall: time dilation of cosmic-ray muons.
1971	Hafele–Keating: atomic clocks flown around the world.

Part B Relativistic Kinematics

Throughout, S and S' are in the standard configuration: S' moves with velocity V along the common x-axis and the origins coincide at $t = t' = 0$. We use

$$\beta = \frac{V}{c}, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}} \quad (\geq 1).$$

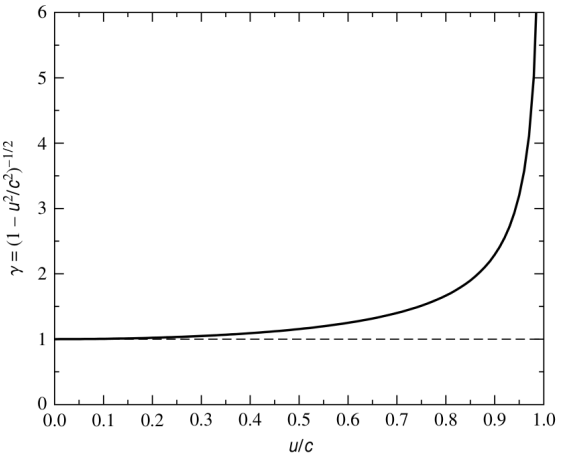


Figure 7: The Lorentz factor γ against $\beta = u/c$. It stays very close to 1 until $\beta \approx 0.3$ and grows without limit as $\beta \to 1$.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics, *Ch. 4, Fig. 5*

β	0.1	0.3	0.5	0.6	0.8	0.9	0.99	0.999
γ	1.005	1.048	1.155	1.25	1.667	2.294	7.09	22.4

Useful approximation for $\beta \ll 1$: $\gamma \approx 1 + \frac{1}{2}\beta^2$.

9 Measuring time in a single frame

Since we are about to discover that time is *not* universal, we must say carefully what "the time of an event in frame S" means (Taylor §15.4). Imagine a chief observer at the origin and a large number of helpers at rest throughout S, each with an accurate clock. The clocks are **synchronised** by light signals: the chief sends a flash at $t = 0$; a helper at distance d sets his clock to d/c when the flash arrives. Any event is then timed by the helper standing right next to it – no signal-delay corrections are needed. In this way every frame assigns a definite (x, y, z, t) to every event.

Important: "observer in S" means this whole team, not one person looking with his eyes. Relativistic effects are *not* optical illusions caused by light travel time.

10 The relativity of simultaneity

Einstein's train thought experiment: a lamp is fixed at the exact middle C of a carriage moving to the right. It flashes. Light travels at c in both directions *in both frames* (Postulate 2).

- **Train frame:** the two ends are equally far from the lamp and at rest, so the light reaches the front and the back *at the same time.*

- **Ground frame:** the back of the train moves *towards* the light and the front moves *away* from it, so the light reaches the *back first*.

Two events that are simultaneous in one frame are not simultaneous in another frame moving relative to it. This is the root of everything that follows.

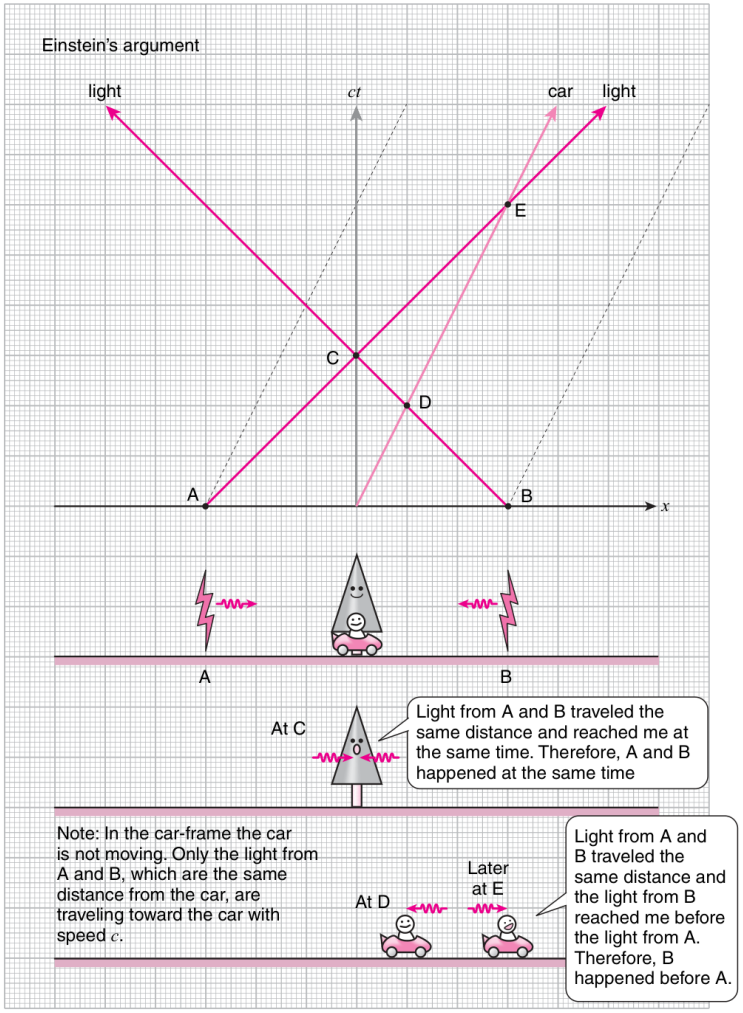


Figure 8: Einstein's argument drawn on a space-time diagram. Light from A and B meets at C; in one frame the emissions at A and B are simultaneous, while for the moving car the light from A and B arrives at different times (D, E), so B is judged to have happened before A.
Source: T. Takeuchi, An Illustrated Guide to Relativity*, §4.5*

Relativity of simultaneity.
If two events are simultaneous in S and separated by Δx along the direction of motion, then in S' they are separated in time by

$$\Delta t' = -\gamma \frac{V\,\Delta x}{c^2}.$$

The event further *forward* (in the direction of S''s motion) happens *earlier* in S'. (Proved from the Lorentz transformation in Section 14.)

11 Space-time diagrams [Unit 3]

This section belongs to Unit 3 (The Geometric Representation of Space-Time). It is placed here because all the kinematic effects of Unit 1 become pictures on a space-time diagram.

A **space-time diagram** (Minkowski diagram) plots events with position x horizontally and ct vertically. Measuring time as ct (a length) gives both axes the same units.

- An **event** is a point.
- The history of a particle is a curve, its **world line**. A body at rest is a vertical line; a body moving uniformly is a straight line with slope $\mathrm{d}(ct)/\mathrm{d}x = c/v$, steeper than 45° since $v < c$.
- A light ray is a line at 45° (slope ± 1). Massive bodies can never have world lines flatter than 45°.
- Lines of simultaneity ("now") are horizontal lines ct = const in this frame; lines of "same place" are vertical lines.

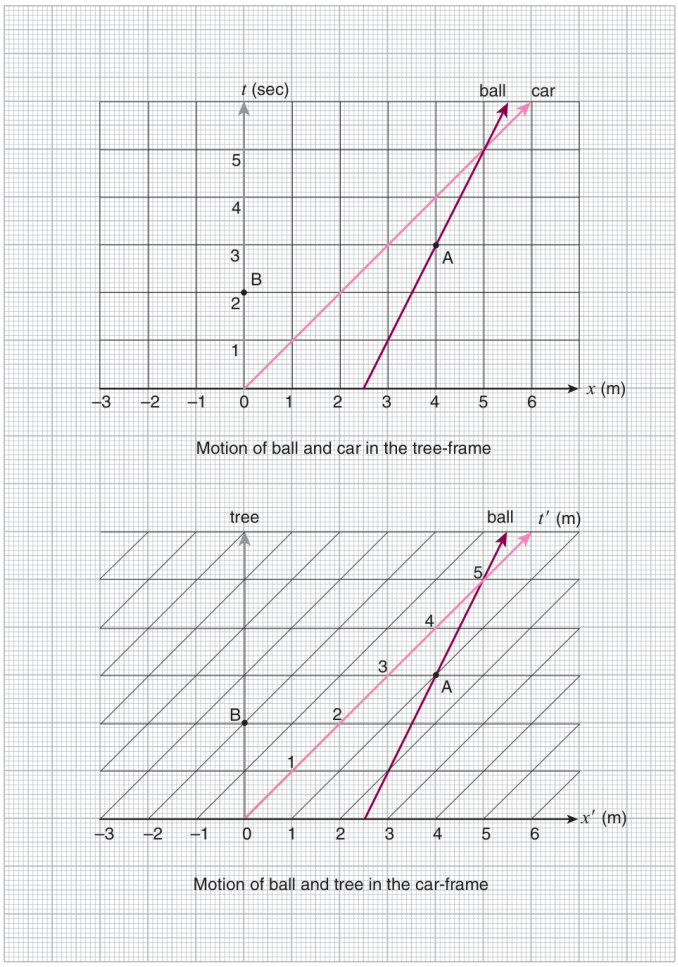


Figure 9: Galilean space-time diagrams. Top: motion of a ball and a car in the tree-frame. Bottom: the same motions in the car-frame. In Galilean physics only the "same place" lines tilt; lines of "same time" stay horizontal in every frame.
Source: T. Takeuchi, An Illustrated Guide to Relativity, *§3.3*

In Newtonian physics, switching frames tilts the time axis (world line of the new origin) but the lines of simultaneity remain horizontal – time is absolute. In relativity, *both* axes tilt (Section 15),

because both “same place” and “same time” depend on the frame. Fig. 10 shows the consequence: clocks synchronised in the car frame are *not* synchronised when viewed from the ground frame, and vice versa.

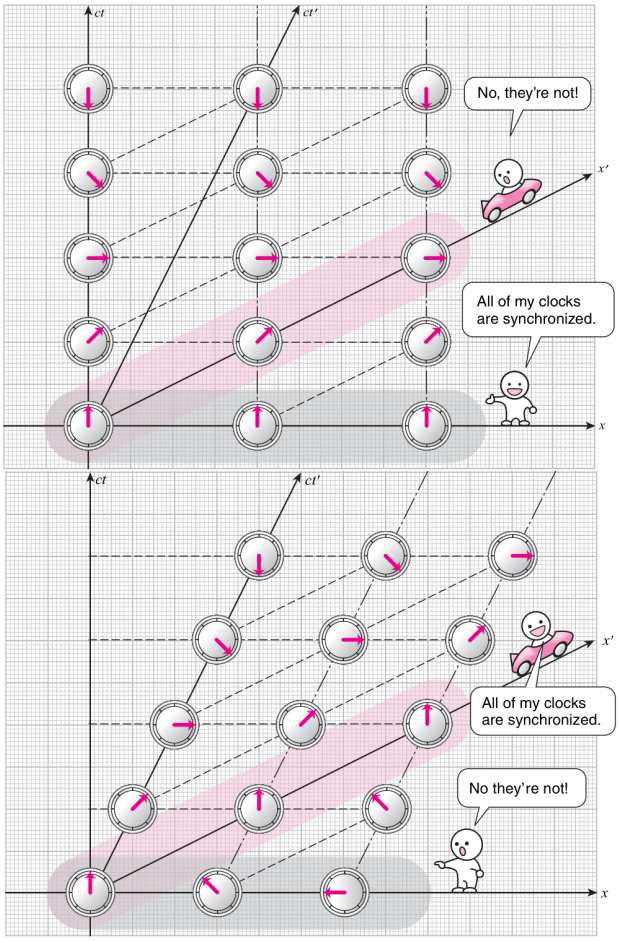


Figure 10: Synchronisation of clocks. Clocks synchronised in one frame (all showing the same reading along a line of simultaneity of that frame) are not synchronised along the line of simultaneity of the other frame.
Source: T. Takeuchi, An Illustrated Guide to Relativity*, §6.1*

12 Time dilation

12.1 Taylor’s light-clock thought experiment

An observer on the train (S') fires a flash bulb on the floor. The light goes up to a mirror on the roof at height h and comes back to a photocell beside the bulb, which beeps (Fig. 11).

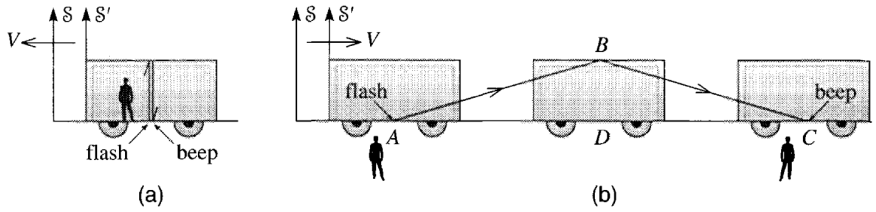


Figure 11: (a) In the train frame S' the light goes straight up and down; flash and beep occur at the same place. (b) In the ground frame S the light travels along ABC; flash and beep occur at different places, a distance $V\Delta t$ apart, so two ground observers are needed to time them.
Source: J. R. Taylor, Classical Mechanics*, Fig. 15.3*

Derivation [5 marks]: Time dilation from the light clock

In S': the light travels $2h$ at speed c: $\quad \Delta t' = 2h/c$.

In S: during the time Δt between flash and beep the train moves $V\Delta t$. The light travels along two sides of an isosceles triangle with half-base $V\Delta t/2$ and height h (heights perpendicular to motion are the same in both frames). By Postulate 2 its speed is still c, so each side has length $c\Delta t/2$:

$$\left(\frac{c\Delta t}{2}\right)^2 = h^2 + \left(\frac{V\Delta t}{2}\right)^2 \;\Rightarrow\; \Delta t = \frac{2h}{\sqrt{c^2 - V^2}} = \frac{2h}{c}\,\frac{1}{\sqrt{1-\beta^2}}.$$

Comparing, $\Delta t = \gamma\,\Delta t'$. ■

The argument applies to *any* two events at the same place in S' (any clock, any process – heartbeat, decay of a particle), because they could always be timed by such a light clock.

Time dilation and proper time.

The time between two events measured in the frame where they occur *at the same place* is the **proper time** Δt_0 (or $\Delta\tau$). In any other frame

$$\Delta t = \gamma\,\Delta t_0 \;\geq\; \Delta t_0.$$

"A moving clock is observed to run slow." The effect is *mutual*: each observer finds the other's clocks slow, because each is comparing one clock of the other frame with two (synchronised) clocks of his own.

Common mistake.

Always identify which frame measures the proper time: it is the frame in which *one* clock is present at both events. Do not decide by which frame is "really moving" – that question has no meaning in relativity.

Solved Example 3: A jet plane (Taylor Ex. 15.1)

A pilot flying at $V = 300\,\mathrm{m\,s^{-1}}$ fires a flash every hour by his clock. What interval do ground observers measure?

$\beta = 10^{-6}$, so $\gamma \approx 1 + \frac{1}{2}\beta^2 = 1 + 5\times10^{-13}$. $\Delta t = \gamma(1\,\mathrm{h}) = 1\,\mathrm{h} + (5\times10^{-13})(3600\,\mathrm{s}) = 1\,\mathrm{h} + 1.8\,\mathrm{ns}$. Less than 2 ns per hour – no wonder classical physics never noticed.

12.2 Experimental evidence

Cosmic-ray muons. Muons are produced 10–15 km up in the atmosphere when cosmic-ray protons hit air nuclei. A muon at rest decays with mean life $\tau_0 = 2.2\,\mu\text{s}$. Even at speed c, a muon would classically travel only $c\tau_0 \approx 660\,\text{m}$ before decaying, yet muons are abundant at sea level. Rossi and Hall (1941) and later Frisch and Smith (1963, Mt Washington, Fig. 12) showed quantitatively that the muons survive because their lifetime is dilated by γ.

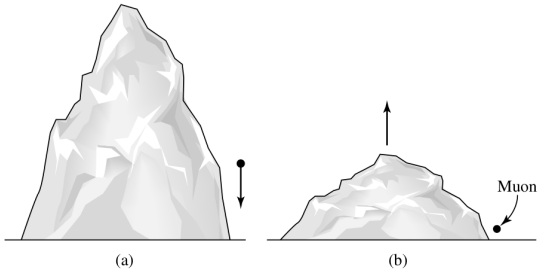


Figure 12: Muons moving down past Mt Washington: (a) in the mountain frame the muon clock runs slow; (b) in the muon frame the mountain is length-contracted. Both descriptions predict the same number of surviving muons.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 4, Fig. 8*

Solved Example 4: Cosmic-ray muons

A muon is created at a height of 10 km and moves straight down at $v = 0.998c$. Mean life at rest $\tau_0 = 2.2\,\mu\text{s}$.

$\gamma = 1/\sqrt{1 - 0.998^2} = 1/\sqrt{0.003996} = 15.8$.

Lifetime in the Earth frame: $\tau = \gamma\tau_0 = 34.8\,\mu\text{s}$. Mean distance travelled: $d = v\tau = (0.998)(3 \times 10^8)(34.8 \times 10^{-6}) \approx 10.4\,\text{km}$.

Classically: $d = v\tau_0 \approx 660\,\text{m}$. Time dilation lets a typical muon reach the ground.

Atomic clocks (Hafele and Keating, 1971). Four caesium clocks were flown round the world on commercial jets, eastward and westward, and compared with clocks at the U.S. Naval Observatory. For the westward trip the predicted gain was (275 ± 21) ns and the observed gain (273 ± 7) ns. (The prediction includes a gravitational part from general relativity – see Section 30.)

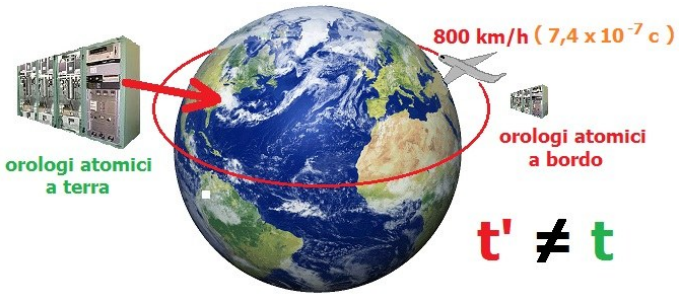


Figure 13: One of the caesium-beam atomic clocks flown around the world in the Hafele–Keating experiment (1971).
Source: Wikimedia Commons, "Hafele–Keating.jpg" (public domain)

Particle accelerators and GPS. Unstable particles moving at $0.99c$ in accelerators live ≈ 7 times longer, exactly as predicted. Global Positioning System satellites carry atomic clocks which must be corrected for both special- and general-relativistic effects (about $38\,\mu\text{s}$ per day); without the correction GPS positions would drift by kilometres per day.

Solved Example 5: Pion beam

Charged pions have proper mean life $\tau_0 = 26\,\text{ns}$. A beam has speed $0.95c$. Find the mean distance travelled before decay.

$\gamma = 1/\sqrt{1-0.9025} = 3.20$; $\tau = \gamma\tau_0 = 83.3\,\text{ns}$; $d = v\tau = (0.95)(3\times10^8)(83.3\times10^{-9}) = 23.7\,\text{m}$ (classically only 7.4 m).

13 Length contraction

Derivation [5 marks]: Length contraction (Taylor §15.5)

Let the train (rest frame S') pass a ground observer Q with speed V.

- In S (ground): Q times the train passing him: Δt. Length $l = V\Delta t$.
- In S' (train): two observers at the front and back record when they pass Q: $\Delta t'$. Length $l' = V\Delta t'$ (the speed of S relative to S' is also V).

The two events "front passes Q" and "back passes Q" occur at the same place in S (at Q), so Δt is the proper time and $\Delta t' = \gamma\Delta t$. Hence

$$l' = V\Delta t' = \gamma V\Delta t = \gamma\, l \qquad \Rightarrow \qquad l = \frac{l'}{\gamma}. \qquad \blacksquare$$

Length contraction.

The length of an object measured in its own rest frame is its **proper length** l_0. In a frame where it moves with speed V along its length,

$$l = \frac{l_0}{\gamma} = l_0\sqrt{1-V^2/c^2} \;\le\; l_0.$$

Lengths *perpendicular* to the motion are unchanged. To measure the length of a moving object one must record the positions of its two ends *simultaneously* in one's own frame.

Why no transverse contraction? (Taylor's knife argument.) Suppose two equally tall observers pass each other, one holding a knife level with his head. If moving bodies shrank sideways, each would see the other shrunk and would expect the knife to pass above (or scalp) the other. The situation is perfectly symmetric, so both cannot be right: there is no transverse contraction.

Solved Example 6: The muon's point of view

For the muon of the previous example, the atmosphere rushes up at $0.998c$. The 10 km of air (proper length, measured on Earth) is contracted to $l = 10\,\text{km}/15.8 = 0.63\,\text{km}$, which the muon covers in $0.63\,\text{km}/(0.998c) = 2.1\,\mu\text{s}$ – less than its proper life. Same conclusion, different explanation: time dilation in one frame is length contraction in the other.

Solved Example 7: A trip to Alpha Centauri

The star is 4.3 light-years (ly) away in the Earth frame. A spaceship travels at $0.8c$ ($\gamma = 5/3$). (a) Earth time for the trip: $4.3/0.8 = 5.38\,\text{yr}$. (b) Ship time (proper time): $5.38/\gamma = 3.23\,\text{yr}$. (c) Distance in the ship frame: $4.3/\gamma = 2.58\,\text{ly}$; check: $2.58/0.8 = 3.23\,\text{yr}$. Consistent.

14 The Lorentz transformation

The Galilean relations $t' = t$ and $x' = x - Vt$ contradict time dilation and length contraction. Taylor derives the correct relations by repeating the reasoning of Fig. 1, now being careful about which frame measures each length.

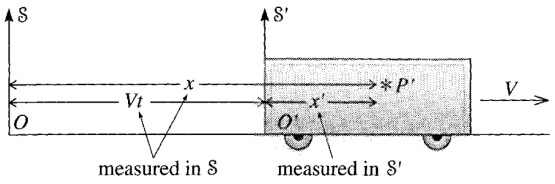


Figure 14: A firecracker explodes and leaves a burn mark at P' on the wall of the carriage. x' is the distance $O'P'$ measured in S'; x and Vt are measured in S at the instant t of the explosion.
Source: J. R. Taylor, Classical Mechanics*, Fig. 15.4*

Derivation [5 marks]: The Lorentz transformation (Taylor §15.6)

(i) Perpendicular lengths are unchanged: $y' = y$, $z' = z$.

(ii) The distance $O'P'$ is x' in S' (a proper length, since the mark is at rest in S'). Measured in S at time t it is $x - Vt$. By length contraction

$$x - Vt = \frac{x'}{\gamma} \quad \Rightarrow \quad x' = \gamma(x - Vt).$$

(iii) Repeating with the roles of the frames exchanged (a mark on a wall fixed in S; S moves with velocity $-V$ relative to S'): $x = \gamma(x' + Vt')$.

(iv) Substitute x' from (ii) into (iii) and solve for t':

$$\frac{x}{\gamma} = \gamma(x - Vt) + Vt' \;\Rightarrow\; Vt' = \frac{x}{\gamma} - \gamma x + \gamma Vt = \gamma Vt - \gamma x\Big(1 - \frac{1}{\gamma^2}\Big) = \gamma Vt - \gamma x\frac{V^2}{c^2}$$

so $t' = \gamma\,(t - Vx/c^2)$. ■

Lorentz transformation (standard configuration).

$$\begin{aligned} x' &= \gamma\,(x - Vt) & x &= \gamma\,(x' + Vt') \\ y' &= y & y &= y' \\ z' &= z & z &= z' \\ t' &= \gamma\Big(t - \frac{Vx}{c^2}\Big) & t &= \gamma\Big(t' + \frac{Vx'}{c^2}\Big) \end{aligned}$$

The inverse is obtained by exchanging primed and unprimed quantities and replacing $V \to -V$. Because the transformation is linear, the same equations hold for *differences*: $\Delta x' = \gamma(\Delta x - V\Delta t)$, $\Delta t' = \gamma(\Delta t - V\Delta x/c^2)$.

Checks.

- *Galilean limit:* for $V \ll c$, $\gamma \to 1$ and $Vx/c^2 \to 0$, giving $x' = x - Vt$, $t' = t$.

- *Light speed is invariant:* a flash from the origin at $t = 0$ has $x = ct$. Then $x' = \gamma(c - V)t$ and $t' = \gamma(1 - V/c)t$, so $x'/t' = c$.
- $V < c$*:* for $V \geq c$, γ is infinite or imaginary. No inertial frame can move at or faster than light relative to another.
- *Matrix form* with $x^0 = ct$:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}.$$

Space and time are mixed together – the starting point of four-dimensional space-time.

14.1 The three effects recovered from the Lorentz transformation

1. **Simultaneity:** two events with $\Delta t = 0$, $\Delta x \neq 0$ in S have $\Delta t' = \gamma(0 - V\Delta x/c^2) = -\gamma V \Delta x/c^2 \neq 0$.
2. **Time dilation:** a clock at rest in S' ($\Delta x' = 0$) ticks $\Delta t' = \Delta t_0$. From the inverse transformation $\Delta t = \gamma(\Delta t' + V\Delta x'/c^2) = \gamma \Delta t_0$.
3. **Length contraction:** a rod at rest in S' has $\Delta x' = l_0$. Its ends are marked in S at the same time ($\Delta t = 0$): $\Delta x' = \gamma(\Delta x - 0) \Rightarrow l = \Delta x = l_0/\gamma$.

Solved Example 8: Transforming an event

In S an event occurs at $x = 900\,\text{m}$, $t = 2.0\,\mu\text{s}$. Frame S' moves at $V = 0.6c$ ($\gamma = 1.25$).

$$x' = 1.25\,[900 - (1.8 \times 10^8)(2.0 \times 10^{-6})] = 1.25\,(900 - 360) = 675\ \text{m},$$

$$t' = 1.25\Big[2.0 \times 10^{-6} - \frac{(1.8 \times 10^8)(900)}{9 \times 10^{16}}\Big] = 1.25\,(2.0 - 1.8)\,\mu\text{s} = 0.25\,\mu\text{s}.$$

Check (see Section 17): $x^2 - c^2t^2 = 900^2 - 600^2 = 4.5 \times 10^5\,\text{m}^2$ and $x'^2 - c^2t'^2 = 675^2 - 75^2 = 4.5 \times 10^5\,\text{m}^2$. The same!

Solved Example 9: A relativistic snake (Taylor Ex. 15.3)

A snake of proper length 100 cm slides past at $V = 0.6c$. A student holds two cleavers 100 cm apart and bounces them on the table *simultaneously* (in his frame S) at $t = 0$, the left one just behind the snake's tail. Student: "The snake is contracted to $100/1.25 = 80$ cm, so the right cleaver misses it." Snake: "The cleavers are contracted to 80 cm apart, so I shall be cut!" Who is right?

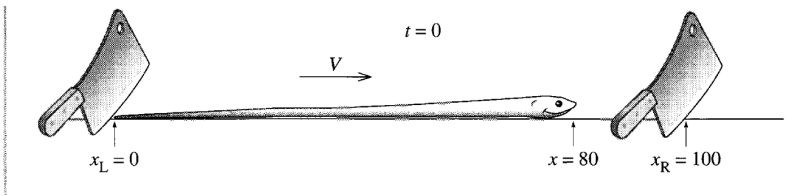


Figure 15: The snake paradox in the student's frame S: the cleavers fall simultaneously at $t = 0$, 100 cm apart; the 80 cm snake is unharmed.
Source: J. R. Taylor, Classical Mechanics*, Fig. 15.5*

Left cleaver: $x_L = 0,\ t_L = 0 \Rightarrow t'_L = 0,\ x'_L = 0$ (just behind the tail). Right cleaver: $x_R = 100\,\text{cm}$, $t_R = 0$:

$$t'_R = \gamma\Big(0 - \frac{Vx_R}{c^2}\Big) = -1.25\,\frac{(0.6)(1\,\text{m})}{3\times10^8\,\text{m/s}} = -2.5\,\text{ns}, \qquad x'_R = \gamma(x_R - 0) = 125\,\text{cm}.$$

In the snake's frame the right cleaver falls *2.5 ns earlier*, at $x' = 125\,\text{cm}$, well beyond the head at 100 cm. Both agree the snake is unhurt. The "paradox" disappears once we remember the relativity of simultaneity.

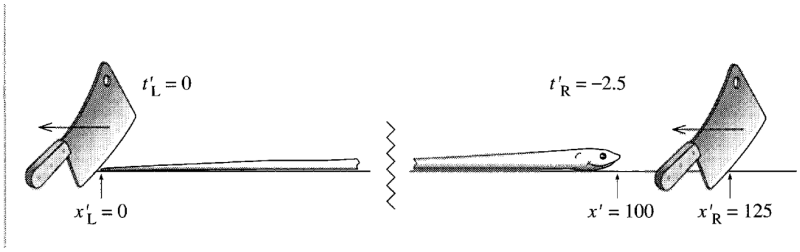


Figure 16: The same events in the snake's frame S': the right cleaver falls 2.5 ns before the left one, so although the cleavers are only 80 cm apart they land 125 cm apart.
Source: J. R. Taylor, Classical Mechanics*, Fig. 15.6*

15 Simultaneity, time dilation and length contraction on a space-time diagram [Unit 3]

Unit 3 topic "Simultaneity, length contraction and time dilation" – the same physics as Sections 10–14, now read off from geometry.

15.1 The axes of a moving frame

Draw the (x, ct) axes of S at right angles. Where are the axes of S'?

- The **ct'-axis** is the world line of the origin of S': $x' = 0 \Rightarrow x = Vt$, i.e. $ct = x/\beta$. It is tilted from the ct-axis towards the light line, by angle α with $\tan\alpha = \beta$.
- The **x'-axis** is the line of events simultaneous with the origin in S': $t' = 0 \Rightarrow ct = \beta x$. It is tilted from the x-axis towards the light line by the *same* angle α.
- The 45° light line bisects both pairs of axes: light has speed c in both frames.

Lines parallel to the x'-axis are lines of simultaneity for S'; they are *not* horizontal. So events on a horizontal line (simultaneous in S) lie on different S'-simultaneity lines – the relativity of simultaneity is simply the tilt of the x'-axis.

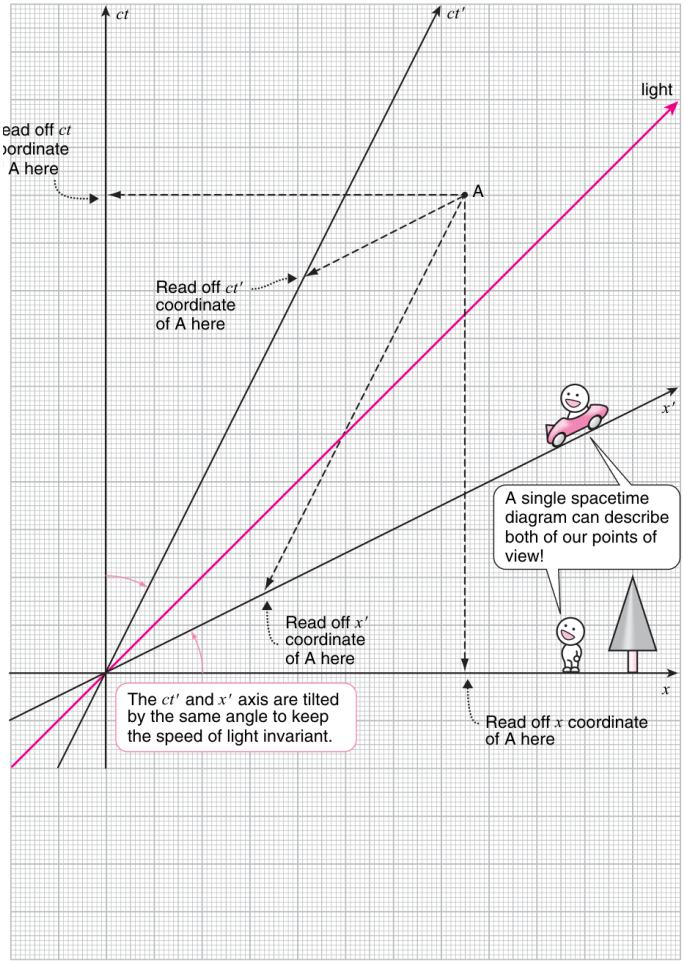


Figure 17: The ct' and x' axes of a moving frame are tilted towards the light line by the same angle, which keeps the speed of light invariant. A single space-time diagram can describe both frames: an event's coordinates in either frame are read off by drawing lines parallel to the corresponding axes.

Source: T. Takeuchi, An Illustrated Guide to Relativity*, §4.8*

Calibration. The unit of length on the tilted axes is *not* the same as on the original axes. The correct scale is given by the invariant hyperbolae $x^2 - c^2t^2 = \pm 1$ (because $x^2 - c^2t^2$ is the same in all frames, Section 17): the point where the hyperbola $c^2t^2 - x^2 = 1$ cuts the ct'-axis is one unit of ct'.

15.2 Time dilation and length contraction as geometry

- **Time dilation.** A clock at rest at the origin of S' ticks along the ct'-axis. The S-observer compares its reading with the S-clock that is level with it on a *horizontal* line; the S'-observer compares using *tilted* lines. Each finds the other's clock behind his own (Fig. 18): the effect is mutual, with no contradiction, because the two observers mean different things by "at the same time".

- **Length contraction.** The world lines of the two ends of a rod at rest in S' form a strip parallel to the ct'-axis. The length in S is the width of the strip along a horizontal line; the length in S' is its width along a tilted line. Measured with the proper calibration, the horizontal cut is shorter by $1/\gamma$ (Fig. 19).

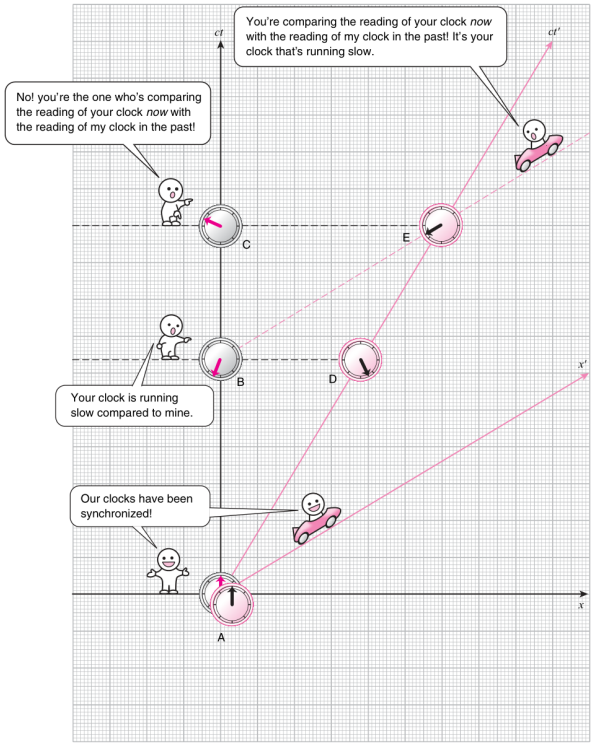


Figure 18: Time dilation on a space-time diagram: each observer, comparing the other's clock with the reading of his own clock "at the same time", finds the other clock running slow.
Source: T. Takeuchi, An Illustrated Guide to Relativity*, §6.2*

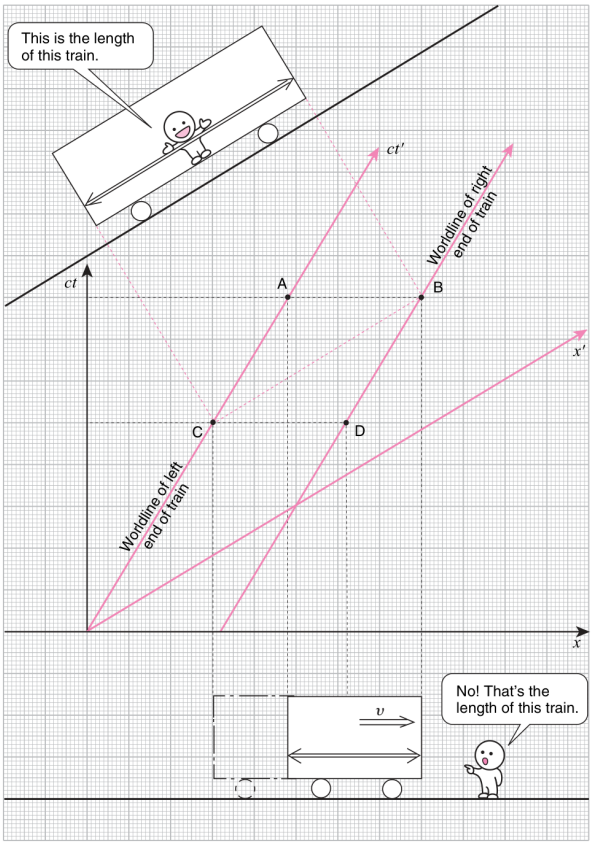


Figure 19: Lorentz contraction: the length of the train is the width of the strip between the world lines of its ends, cut along each observer's own line of simultaneity.
Source: T. Takeuchi, An Illustrated Guide to Relativity*, §6.4*

What time dilation and length contraction do *not* mean.
Time does not "flow more slowly" in some absolute sense, and space is not physically squeezed (this was Lorentz's view, not Einstein's). Both effects are statements about how two observers, who disagree about simultaneity, compare measurements (Takeuchi §§6.3, 6.5).

16 The relativistic velocity-addition formula

A particle has velocity $\mathbf{v} = (v_x, v_y, v_z)$ in S. What is its velocity $\mathbf{v}'$ in S'?

Derivation [5 marks]: Velocity transformation (Taylor §15.7)
For two neighbouring points on the particle's path, the differences obey

$$\mathrm{d}x' = \gamma(\mathrm{d}x - V\,\mathrm{d}t), \quad \mathrm{d}y' = \mathrm{d}y, \quad \mathrm{d}z' = \mathrm{d}z, \quad \mathrm{d}t' = \gamma\Big(\mathrm{d}t - \frac{V\,\mathrm{d}x}{c^2}\Big).$$

Divide, then divide numerator and denominator by $\mathrm{d}t$:

$$v'_x = \frac{\mathrm{d}x'}{\mathrm{d}t'} = \frac{\gamma(\mathrm{d}x - V\,\mathrm{d}t)}{\gamma(\mathrm{d}t - V\,\mathrm{d}x/c^2)} = \frac{v_x - V}{1 - v_x V/c^2}, \qquad v'_y = \frac{\mathrm{d}y'}{\mathrm{d}t'} = \frac{v_y}{\gamma(1 - v_x V/c^2)},$$

and similarly $v'_z = v_z/[\gamma(1 - v_x V/c^2)]$. ■

Relativistic velocity addition.

$$v'_x = \frac{v_x - V}{1 - v_x V/c^2}, \qquad v'_{y,z} = \frac{v_{y,z}}{\gamma\,(1 - v_x V/c^2)}; \qquad \text{inverse: } v_x = \frac{v'_x + V}{1 + v'_x V/c^2}.$$

For $v, V \ll c$ these reduce to $v' = v - V$. If $v'_x = c$ then $v_x = c$: light has speed c in all frames. If $v' < c$ and $V < c$ then $v < c$: velocities can never be "added" to exceed c.

Solved Example 10: Adding velocities close to c (Taylor Ex. 15.4, 15.5)

A rocket moving at $0.8c$ relative to Earth fires bullets forward at $0.6c$ relative to itself.

$$v = \frac{0.6c + 0.8c}{1 + (0.6)(0.8)} = \frac{1.4c}{1.48} = 0.946c \quad (\text{not } 1.4c).$$

If instead it fires a light pulse ($v' = c$): $v = (c + 0.8c)/(1 + 0.8) = c$.

Solved Example 11: Two spaceships approaching

Ships A and B approach the Earth from opposite sides, each at $0.9c$. Velocity of B relative to A: take S' = rest frame of A ($V = +0.9c$), $v_x = -0.9c$:

$$v'_x = \frac{-0.9c - 0.9c}{1 - (-0.9)(0.9)} = \frac{-1.8c}{1.81} = -0.9945c.$$

Solved Example 12: Fizeau's experiment explained

Light moves in water (refractive index n) with speed c/n relative to the water; the water flows at $u \ll c$. In the laboratory

$$v = \frac{c/n + u}{1 + u/(nc)} \approx \Big(\frac{c}{n} + u\Big)\Big(1 - \frac{u}{nc}\Big) \approx \frac{c}{n} + u\Big(1 - \frac{1}{n^2}\Big).$$

This is exactly Fresnel's "drag coefficient" $(1 - 1/n^2)$ measured by Fizeau in 1851 – no ether needed.

17 Four-dimensional space-time, the interval s^2 and four-vectors [Unit 3]

Unit 3: the geometric representation of space-time. Minkowski (1908) realised that the Lorentz transformation is a kind of "rotation" in a four-dimensional space-time. Here we build the basic tools – the invariant interval and four-vectors – and introduce just enough tensor language to use them.

17.1 An analogy: rotations in ordinary space

Rotate the x, y axes by an angle ϕ. The coordinates of a point change, $x' = x\cos\phi + y\sin\phi$, $y' = -x\sin\phi + y\cos\phi$, but its distance from the origin does not: $x'^2 + y'^2 = x^2 + y^2$. Any set of three numbers that transforms like (x, y, z) under rotations is a **three-vector**, and its length (and the dot product of two such vectors) is a **scalar** – the same for all orientations of the axes.

17.2 The space-time interval

For two events separated by $(\Delta x, \Delta y, \Delta z, \Delta t)$ define the **interval**

The invariant interval (sign convention of Taylor and Griffiths).

$$s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2 = \Delta r^2 - c^2\Delta t^2.$$

s^2 has the *same value in every inertial frame*: it is a Lorentz scalar (invariant).

Derivation [5 marks]: Invariance of the interval
Under the standard boost $\Delta x' = \gamma(\Delta x - \beta c\Delta t)$, $c\Delta t' = \gamma(c\Delta t - \beta\Delta x)$, $\Delta y' = \Delta y$, $\Delta z' = \Delta z$:

$$\Delta x'^2 - c^2\Delta t'^2 = \gamma^2\big[(\Delta x - \beta c\Delta t)^2 - (c\Delta t - \beta\Delta x)^2\big] = \gamma^2\big[\Delta x^2(1-\beta^2) - c^2\Delta t^2(1-\beta^2)\big]$$

(the cross terms $\mp 2\beta\Delta x\, c\Delta t$ cancel). Since $\gamma^2(1-\beta^2) = 1$,

$$\Delta x'^2 + \Delta y'^2 + \Delta z'^2 - c^2\Delta t'^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2, \quad \text{i.e. } s'^2 = s^2. \qquad \blacksquare$$

Sign conventions.
Narlikar and Carroll–Ostlie write the interval with the opposite overall sign, $\Delta s^2 = c^2\Delta t^2 - \Delta r^2$. Nothing physical changes; only the words "positive" and "negative" swap. In these notes (and in the assignments, unless stated otherwise) we use $s^2 = \Delta r^2 - c^2\Delta t^2$.

Physical meaning. A flash at the origin spreads as $r = ct$, i.e. $s^2 = 0$. Since s^2 is invariant, it is zero in every frame – the light sphere is a light sphere for everyone. The invariance of s^2 is Postulate 2 written as geometry.

Proper time and proper distance.

- If $s^2 < 0$ there is a frame where the two events are at the same place. The time between them in that frame is the **proper time** $\Delta\tau = \sqrt{-s^2}/c = \sqrt{\Delta t^2 - \Delta r^2/c^2}$.
- If $s^2 > 0$ there is a frame where they are simultaneous. Their separation in that frame is the **proper distance** $\sqrt{s^2}$.

For a moving clock, $\mathrm{d}\tau = \sqrt{\mathrm{d}t^2 - \mathrm{d}r^2/c^2} = \mathrm{d}t\sqrt{1 - v^2/c^2} = \mathrm{d}t/\gamma$ – time dilation follows directly from the invariance of s^2.

17.3 Four-vectors

Label the coordinates of an event by an upper index $\mu = 0, 1, 2, 3$:

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct,\, x,\, y,\, z).$$

(Taylor puts time last, $x_4 = ct$; Griffiths and most modern books put it first. We follow Griffiths so that Units 1, 2 and 3 use the same notation.) The Lorentz transformation is written

$$x'^\mu = \sum_{\nu=0}^{3} \Lambda^\mu{}_\nu\, x^\nu, \qquad \Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Definition. A **four-vector** is any set of four quantities $a^\mu = (a^0, a^1, a^2, a^3)$ that transforms between inertial frames exactly as x^μ does: $a'^\mu = \Lambda^\mu{}_\nu a^\nu$.

Scalar product. For two four-vectors

$$a \cdot b = -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3 = \mathbf{a} \cdot \mathbf{b} - a^0 b^0,$$

which is invariant (the proof is the same as for s^2). In particular $a \cdot a$ is invariant, and $s^2 = \Delta x \cdot \Delta x$. Note that $a \cdot a$ can be positive, negative or zero.

17.4 A first look at tensors

Tensors are the natural language for writing laws that hold in every inertial frame.

1. **Summation convention.** A Greek index repeated once up and once down is summed from 0 to 3: $\Lambda^\mu{}_\nu x^\nu \equiv \sum_\nu \Lambda^\mu{}_\nu x^\nu$.

2. **The metric tensor.** $g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$, so that $s^2 = g_{\mu\nu}\Delta x^\mu \Delta x^\nu$ and $a \cdot b = g_{\mu\nu} a^\mu b^\nu$.

3. **Contravariant and covariant components.** A vector with an upper index a^μ is *contravariant*. Lowering the index with the metric gives the *covariant* components

   $$a_\mu = g_{\mu\nu} a^\nu = (-a^0, a^1, a^2, a^3).$$

   Then $a \cdot b = a_\mu b^\mu$ – the minus sign is hidden in the lowered index.

4. **Rank (order).** A *scalar* (rank 0) has one component, the same in all frames (c, s^2, τ, rest mass m, charge q). A *four-vector* (rank 1) has 4 components and one index. A *rank-2 tensor* has 16 components $T^{\mu\nu}$, transforming with one Λ for each index:

   $$T'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta \, T^{\alpha\beta}.$$

   Examples: the metric $g_{\mu\nu}$; the outer product $a^\mu b^\nu$; the electromagnetic field tensor $F^{\mu\nu}$ (Unit 2), whose six independent components are $\mathbf{E}$ and $\mathbf{B}$.

5. **Symmetric and antisymmetric tensors:** $T^{\mu\nu} = T^{\nu\mu}$ (e.g. $g_{\mu\nu}$) or $T^{\mu\nu} = -T^{\nu\mu}$ (e.g. $F^{\mu\nu}$; its diagonal is zero and it has only 6 independent components). These properties are the same in every frame.

6. **Contraction.** Summing an upper with a lower index lowers the rank by two: $a_\mu b^\mu$ is a scalar; $T^\mu{}_\mu$ is a scalar.

7. **Quotient rule** (Taylor §15.11). If $k_\mu x^\mu$ is a scalar for every four-vector x^μ, then k^μ is a four-vector. (Used below for the wave four-vector.)

Why bother with tensors?.
If a law of physics is written as "tensor = tensor" (e.g. $p^\mu_{\text{before}} = p^\mu_{\text{after}}$, or $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$ in Unit 2), both sides transform with the same Λ's. So if it holds in one inertial frame it holds in all of them – Postulate 1 is automatically satisfied. This is called **covariance**. It is the same reason that $\mathbf{F} = m\mathbf{a}$, written in vectors, is true whichever way we orient our axes.

Solved Example 13: Working with a four-vector
A four-vector has components $a^\mu = (5, 3, 0, 0)$ in S. Find (a) a_μ, (b) $a \cdot a$, (c) its components in S' moving with $\beta = 0.6$ along x, and check (b).

(a) $a_\mu = (-5, 3, 0, 0)$. (b) $a \cdot a = a_\mu a^\mu = -25 + 9 = -16$ (negative: "timelike").

(c) $\gamma = 1.25$: $a'^0 = \gamma(a^0 - \beta a^1) = 1.25(5 - 1.8) = 4$, $a'^1 = \gamma(a^1 - \beta a^0) = 1.25(3 - 3) = 0$. So $a'^\mu = (4, 0, 0, 0)$ and $a' \cdot a' = -16$. ✓ (In S' the vector is "pure time".)

Solved Example 14: Scalar product of two four-vectors
$a^\mu = (2, 1, 1, 0)$, $b^\mu = (3, 2, 0, 1)$. Then $a \cdot b = -6 + 2 + 0 + 0 = -4$, $a \cdot a = -4 + 1 + 1 = -2$, $b \cdot b = -9 + 4 + 1 = -4$. These numbers are the same in every inertial frame.

18 The time order and space separation of events [Unit 3]

Unit 3 topic. The sign of s^2, which every observer agrees on, decides whether the time order of two events is absolute or relative – and therefore whether one can cause the other.

18.1 The light cone

Take an event O at the origin. All light rays through O form the **light cone** $r^2 = c^2t^2$ (i.e. $s^2 = 0$). With one space dimension suppressed it is a double cone with 45° sides (Fig. 20). Because s^2 is invariant, all observers agree on the light cone. It divides space-time into regions:

- **Absolute future** (inside the upper cone): $s^2 < 0$, $t > 0$.
- **Absolute past** (inside the lower cone): $s^2 < 0$, $t < 0$.
- **Elsewhere** (outside the cone): $s^2 > 0$.
- The cone itself (forward and backward light cones): $s^2 = 0$.

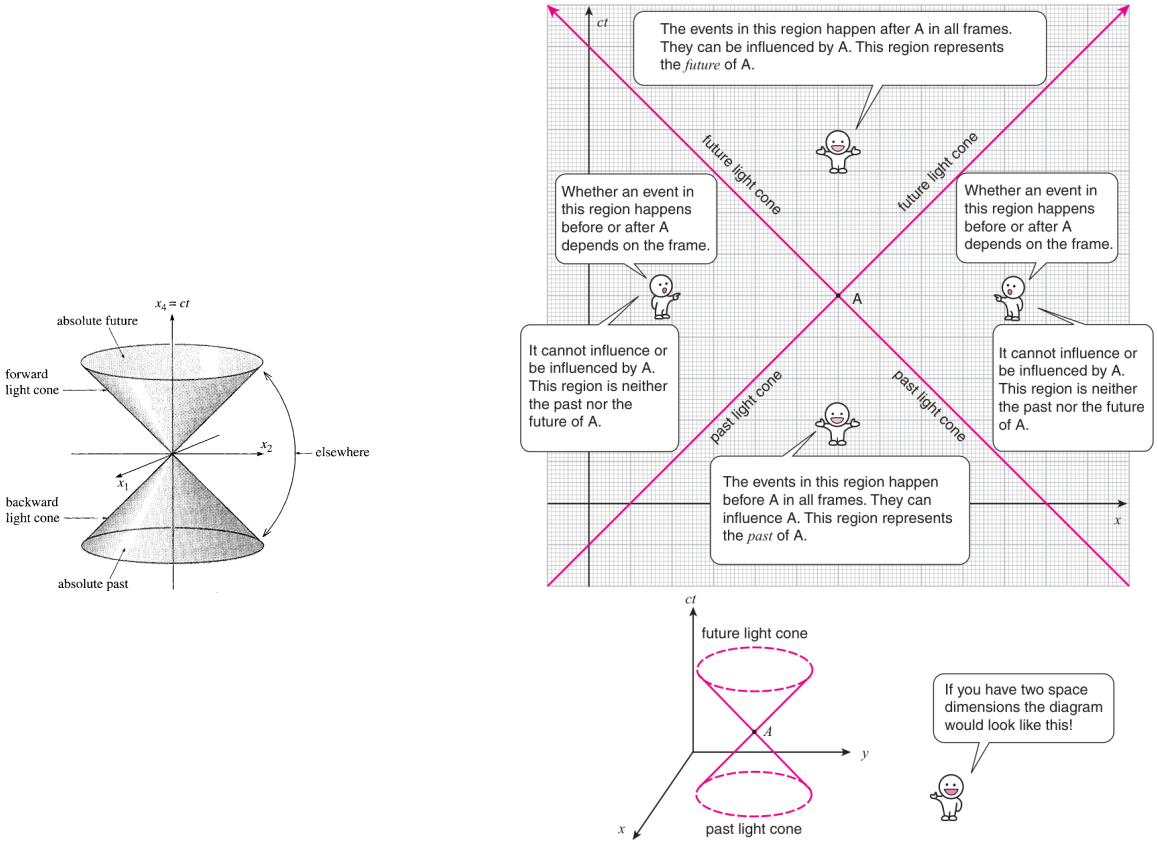


Figure 20: **Left:** the light cone $x \cdot x = r^2 - c^2t^2 = 0$ divides space-time into the forward and backward light cones, the absolute future and absolute past, and "elsewhere".
Source: J. R. Taylor, Classical Mechanics, *Fig. 15.8* **Right:** the same regions on a space-time diagram with the causal relations to event A written in.
Source: T. Takeuchi, An Illustrated Guide to Relativity, *§5.5*

18.2 Classification of intervals

Timelike, spacelike and lightlike separations.

$s^2 < 0$	**timelike**	$\lvert\Delta r\rvert < c\lvert\Delta t\rvert$. A frame exists in which both events are at the same place. Their time order is the *same in all frames*. One can cause the other (a signal slower than light can connect them). Proper time $\sqrt{-s^2}/c$.
$s^2 = 0$	**lightlike** (null)	Connected exactly by a light signal. Time order absolute.
$s^2 > 0$	**spacelike**	$\lvert\Delta r\rvert > c\lvert\Delta t\rvert$. A frame exists in which they are simultaneous, and others in which their order is *reversed*. Neither can cause the other. Proper distance $\sqrt{s^2}$.

Derivation [5 marks]: The time order of timelike-separated events is absolute

Let event P be at $(\Delta x, \Delta t)$ relative to O, with $\Delta t > 0$ and $|\Delta x| < c\Delta t$ (timelike). In any frame S', $\Delta t' = \gamma(\Delta t - V\Delta x/c^2)$. Since $|V| < c$ and $|\Delta x| < c\Delta t$, we have $|V\Delta x/c^2| < (c)(c\Delta t)/c^2 = \Delta t$, so $\Delta t - V\Delta x/c^2 > 0$ and $\Delta t' > 0$. Every observer agrees that P is later than O.

For a spacelike pair, $|\Delta x| > c\Delta t$, and choosing $V = c^2\Delta t/\Delta x$ (which is $< c$) gives $\Delta t' = 0$; a slightly larger V (still $< c$) gives $\Delta t' < 0$. The order is frame-dependent. ■

Causality and the speed limit.

If a cause A produced an effect B at spacelike separation, some observers would see the effect before its cause. To keep causality in every frame, **no signal or influence can travel faster than light**. Material particles have timelike world lines ($\mathrm{d}x \cdot \mathrm{d}x = (v^2 - c^2)\mathrm{d}t^2 < 0$), so $v < c$ in every frame (Taylor §15.10).

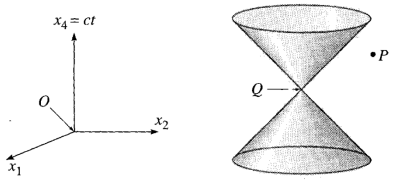


Figure 21: The light cone of an arbitrary event Q. The event P lies outside it: P and Q are spacelike separated, and nothing at Q can influence P.
Source: J. R. Taylor, Classical Mechanics*, Fig. 15.9*

Solved Example 15: A timelike pair

Event A: $(x, ct) = (0, 0)$; event B: $x = 3\,\mathrm{ly}$, $ct = 5\,\mathrm{ly}$ (i.e. $t = 5\,\mathrm{yr}$).

$$s^2 = 3^2 - 5^2 = -16\ \mathrm{ly}^2 < 0 \ \Rightarrow\ \text{timelike.}$$

Proper time $\Delta\tau = \sqrt{16} = 4\,\mathrm{yr}$. A frame moving with $\beta = \Delta x/(c\Delta t) = 3/5$ sees both events at the same place; there $\gamma = 1.25$ and $c\Delta t' = \gamma(5 - 0.6 \times 3) = 4\,\mathrm{ly}$, $\Delta x' = \gamma(3 - 0.6 \times 5) = 0$. A is before B in every frame; A can cause B.

Solved Example 16: A spacelike pair

Event A at the origin; event B at $x = 5\,\mathrm{ly}$, $ct = 3\,\mathrm{ly}$. $s^2 = 25 - 9 = +16\,\mathrm{ly}^2 > 0$: spacelike, proper distance $4\,\mathrm{ly}$.

- $\beta = c\Delta t/\Delta x = 0.6$: $c\Delta t' = 1.25(3 - 0.6\times 5) = 0$ (simultaneous), $\Delta x' = 1.25(5-1.8) = 4\,\text{ly}$.
- $\beta = 0.8$ ($\gamma = 5/3$): $c\Delta t' = \frac{5}{3}(3-4) = -1.67\,\text{ly}$: now B happens *before* A.

Neither event can influence the other.

Solved Example 17: Can a supernova cause it?

A star explodes 20 ly from Earth. A detector on Earth records a burst of particles 15 yr (Earth time) after the explosion. Could the explosion have caused the burst?

$s^2 = 20^2 - 15^2 = 175\,\text{ly}^2 > 0$: spacelike. The signal would have to travel at $20/15 = 1.33c$. **No.**

19 The twin paradox [Unit 3]

> *Unit 3 topic. The space-time interval along a world line is the time recorded by a clock carried along it – this resolves the most famous "paradox" of relativity.*

Stella flies to a star and back at high speed while her twin Terence stays on Earth. Terence says: "Her clock runs slow; she will return younger." But from Stella's ship, the Earth moves away and back, so shouldn't *Terence's* clock run slow? Who is younger?

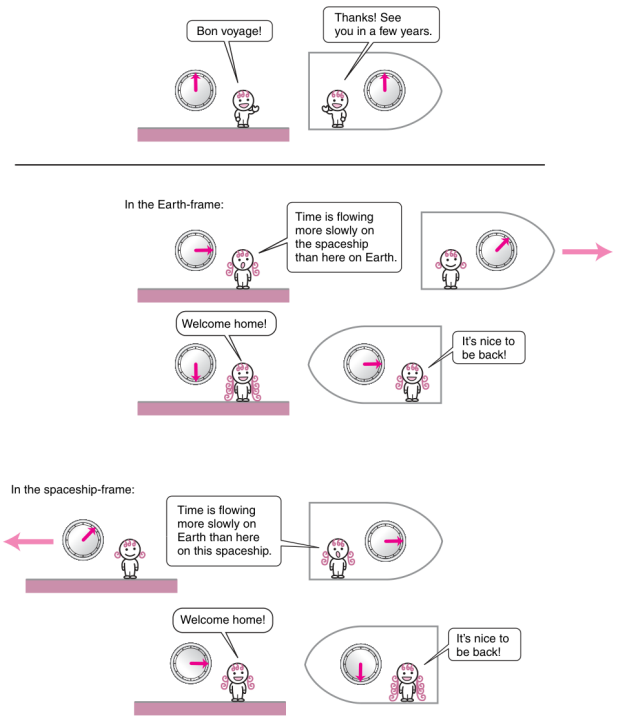


Figure 22: The twin paradox: each twin, reasoning in his or her own frame, expects the other to be younger on reunion. Which expectation is correct?

Source: T. Takeuchi, An Illustrated Guide to Relativity*, §6.6*

Resolution. The situation is *not symmetric*. Terence stays in one inertial frame throughout. Stella does not: she must turn round, i.e. change from the outbound inertial frame to the inbound one (she feels the acceleration; Terence feels nothing). The time-dilation formula may be applied by Terence (an inertial observer) but not naively by Stella across the turn-around.

In space-time language: the time shown by a clock is the proper time along its world line,

$$\tau = \int \sqrt{\mathrm{d}t^2 - \mathrm{d}r^2/c^2} = \int \frac{\mathrm{d}t}{\gamma}.$$

The straight (inertial) world line between two events has the *longest* proper time; any bent world line between the same events has less (the minus sign in s^2 reverses the usual triangle inequality). So Stella, whose world line is bent, is younger.

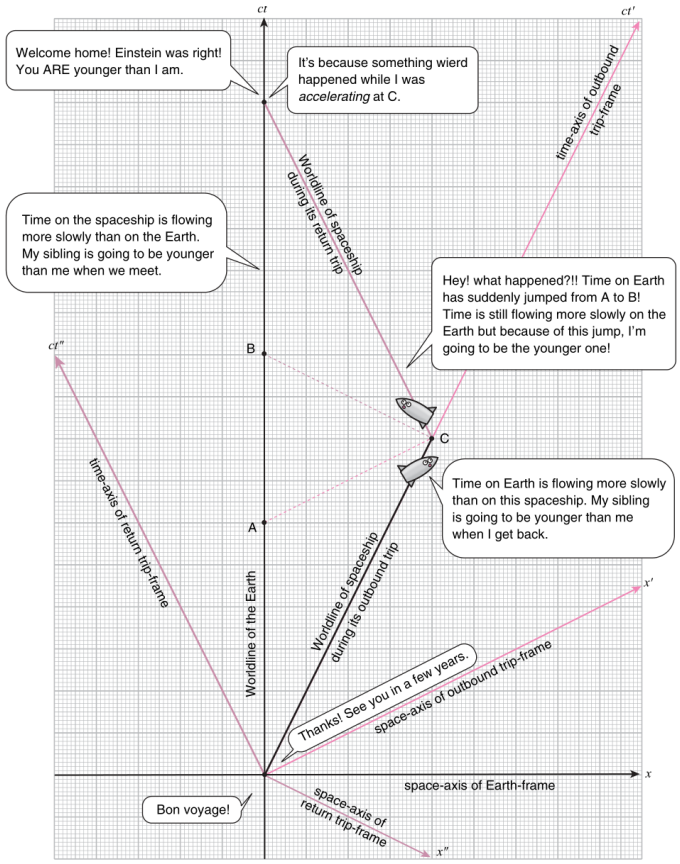


Figure 23: Space-time diagram of the twin paradox. At the turn-around C the spaceship's line of simultaneity swings suddenly from CA to CB: in the ship's view the Earth clock "jumps" forward. This accounts for all the extra ageing of the stay-at-home twin.
Source: T. Takeuchi, An Illustrated Guide to Relativity*, §6.6*

Solved Example 18: The twins

Stella travels to a star 4 ly away (Earth frame) at $0.8c$ ($\gamma = 5/3$) and returns at the same speed.

- **Earth (Terence):** each leg takes $4/0.8 = 5$ yr; total **10 yr**.
- **Stella:** each leg has $\Delta x = 4$ ly, $c\Delta t = 5$ ly, so $c\Delta\tau = \sqrt{5^2 - 4^2} = 3$ ly, i.e. 3 yr. Total **6 yr**.

Stella returns 4 years younger than her twin.

Check with signals: each twin sends a light signal every year. The Doppler factor (Section 21) for receding at $0.8c$ is $\sqrt{(1-\beta)/(1+\beta)} = 1/3$, for approaching it is 3. On the way out (3 yr of her time) Stella receives Earth signals at $1/3$ per year: 1 signal; on the way back (3 yr) at 3 per year: 9 signals. Total 10 – Terence has aged 10 years. Both agree.

The twin effect has been tested directly: muons circulating in the CERN storage ring (1977) lived longer by exactly $\gamma = 29.3$, even though their motion is circular (accelerated), and the Hafele–Keating clocks returned showing less elapsed time than clocks that stayed behind (after the gravitational correction).

20 Aberration of light

Aberration is the change in the *direction* of a light ray between two frames (Bradley's stellar aberration was met in Part A).

Derivation [5 marks]: Relativistic aberration formula

In the source frame S' a light ray makes angle θ' with the x'-axis: $v'_x = c\cos\theta'$, $v'_y = c\sin\theta'$. Using the inverse velocity transformation ($v_x = (v'_x + V)/(1 + v'_x V/c^2)$, $v_y = v'_y/[\gamma(1 + v'_x V/c^2)]$), and knowing that the speed in S is still c:

$$\cos\theta = \frac{v_x}{c} = \frac{\cos\theta' + \beta}{1 + \beta\cos\theta'}, \qquad \sin\theta = \frac{v_y}{c} = \frac{\sin\theta'}{\gamma(1 + \beta\cos\theta')}.$$

Inverting ($\beta \to -\beta$): $\cos\theta' = \dfrac{\cos\theta - \beta}{1 - \beta\cos\theta}$. ■

(a) Stellar aberration. A star lies at $\theta = 90°$ in the Sun's frame S. The Earth (S') moves along x with $\beta = 10^{-4}$. Then $\cos\theta' = -\beta$, so the star appears tilted forward by $\alpha = \theta' - 90°$ with $\sin\alpha = \beta$, i.e. $\alpha \approx \beta = 20.6''$ – Bradley's result, now exact for any speed.

(b) Headlight effect. Light emitted at $\theta' = 90°$ in the source frame is seen in the lab at $\cos\theta = \beta$, i.e. $\sin\theta = 1/\gamma$. Half of all the light emitted by a fast source is concentrated into a forward cone of half-angle $\approx 1/\gamma$ (Fig. 24). This *relativistic beaming* explains the narrow beams of synchrotron radiation and the one-sided jets of quasars (Unit 3).

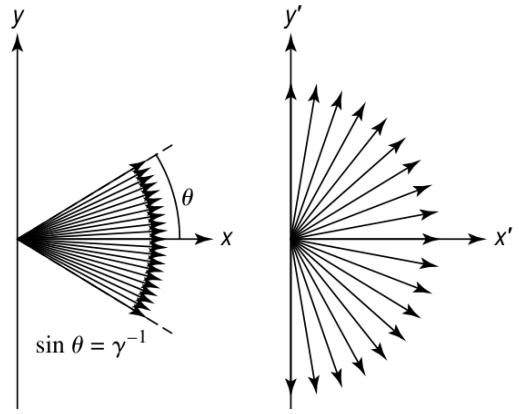


Figure 24: The relativistic headlight effect. Light emitted uniformly in the source frame (right) is concentrated in the forward direction in the frame where the source moves fast (left); half of it lies within $\sin\theta = \gamma^{-1}$.

Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 4, Fig. 10*

Solved Example 19: Headlight cone

For $\beta = 0.6$: $\cos\theta = 0.6$, $\theta = 53.1°$. For an electron in a synchrotron with $\gamma = 1000$: $\theta \approx 1/\gamma = 10^{-3}\,\text{rad} = 0.06°$.

21 The relativistic Doppler effect

For sound, the Doppler shift depends on whether the source or the observer moves relative to the air. Light has no medium, so only the *relative* velocity can matter – and time dilation now enters.

Derivation [5 marks]: Longitudinal Doppler effect (source receding)

A source moves directly away from an observer with speed V and emits crests with period T_0 in its own frame (frequency $f_0 = 1/T_0$).

(i) In the observer's frame the source's clock is slow: crests are emitted every γT_0.

(ii) In that time the source moves a further $V\gamma T_0$ away, so each crest has extra distance $V\gamma T_0$ to travel, taking extra time $\beta\gamma T_0$.

(iii) Observed period: $T = \gamma T_0(1+\beta) = T_0 \dfrac{1+\beta}{\sqrt{(1-\beta)(1+\beta)}} = T_0\sqrt{\dfrac{1+\beta}{1-\beta}}$.

Hence $f = f_0\sqrt{\dfrac{1-\beta}{1+\beta}}$ (receding); for an approaching source $\beta \to -\beta$. ■

21.1 The general formula (Taylor §15.11)

A plane wave $\cos(\mathbf{k}\cdot\mathbf{r} - \omega t)$ has phase $\mathbf{k}\cdot\mathbf{r} - \omega t = k_\mu x^\mu$ with $k^\mu = (\omega/c, \mathbf{k})$. The phase (counting crests) is the same in all frames, so by the quotient rule k^μ is a four-vector – the **wave four-vector**. Transforming its time component from the lab to the source frame, $\omega'/c = \gamma(\omega/c - \beta k_x)$ with $k_x = (\omega/c)\cos\theta$, gives

Relativistic Doppler formula.

$$\omega = \frac{\omega_0}{\gamma\,(1 - \beta\cos\theta)},$$

where ω_0 is the frequency in the source frame and θ is the angle (in the observer's frame) between the source velocity and the line from source to observer.

- Approaching ($\theta = 0$): $f = f_0\sqrt{\dfrac{1+\beta}{1-\beta}}$ (blue shift)
- Receding ($\theta = \pi$): $f = f_0\sqrt{\dfrac{1-\beta}{1+\beta}}$ (red shift)
- Transverse ($\theta = \pi/2$): $f = f_0/\gamma$ (a pure time-dilation red shift, with no classical analogue; first measured by Ives and Stilwell, 1938)

Red shift parameter: $z = \dfrac{\lambda_{\text{obs}} - \lambda_{\text{emit}}}{\lambda_{\text{emit}}}$; for recession $1 + z = \sqrt{\dfrac{1+\beta}{1-\beta}}$, and $z \approx \beta$ when $\beta \ll 1$.

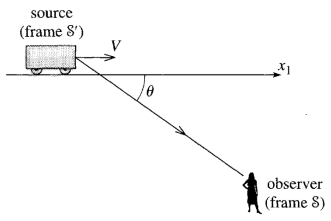


Figure 25: The Doppler experiment: a source moves along x with speed V; the observer in S receives its light at angle θ to the x-axis.

Source: J. R. Taylor, Classical Mechanics, *Fig. 15.10*

Solved Example 20: A receding galaxy

The Hα line ($\lambda_0 = 656.3\,\text{nm}$) from a galaxy is observed at 721.9 nm. Find the speed.

$1 + z = 721.9/656.3 = 1.100$, so $(1+\beta)/(1-\beta) = 1.21 \Rightarrow \beta = 0.21/2.21 = 0.095$: $v = 0.095c = 2.85 \times 10^4\,\text{km}\,\text{s}^{-1}$, receding.

Solved Example 21: The quasar 3C 273

3C 273 has $z = 0.158$. Using the relativistic formula,

$$\beta = \frac{(1+z)^2 - 1}{(1+z)^2 + 1} = \frac{1.341 - 1}{1.341 + 1} = 0.146,$$

i.e. $v \approx 4.4 \times 10^4\,\text{km}\,\text{s}^{-1}$. (For cosmological distances the red shift is really due to the expansion of space – Unit 3 notes, Part B.)

Solved Example 22: Transverse Doppler shift

Atoms moving at $0.01c$ emit light of $\lambda_0 = 500\,\text{nm}$; observed at right angles to their motion, $\lambda = \gamma\lambda_0 = (1 + 5 \times 10^{-5})(500) = 500.025\,\text{nm}$, a shift of 0.025 nm.

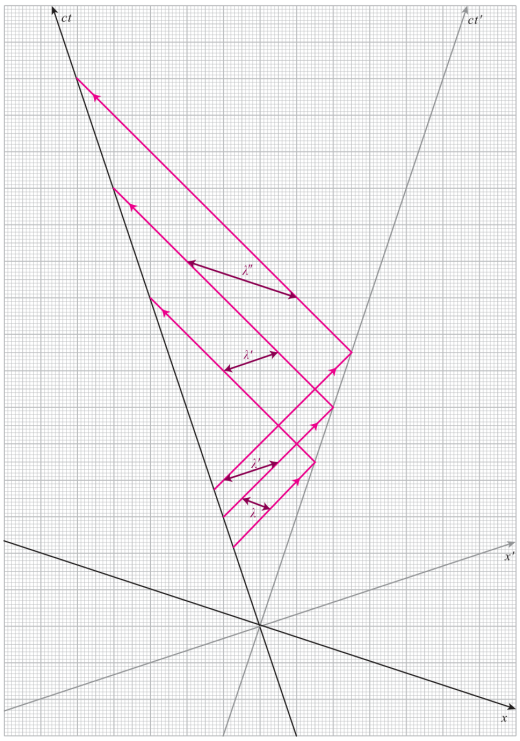


Figure 26: Red shift on a space-time diagram: crests leaving a receding source (pink world lines at 45°) arrive at the observer more widely spaced than they were emitted.

Source: T. Takeuchi, An Illustrated Guide to Relativity*, §6.7*

Part C Relativistic Dynamics

So far we have only described motion. Now we ask what happens to momentum, energy and force. The guiding principle (Taylor §15.12): *conservation laws must hold in every inertial frame* (Postulate 1), and the new laws must reduce to Newton's when $v \ll c$.

22 Why momentum must be redefined

Consider a glancing elastic collision of two identical balls (Fig. 27). In frame S they approach symmetrically and classical momentum $m\mathbf{v}$ is conserved. Now view the same collision from a frame S' moving along x_1. The y-velocities transform as $v'_y = v_y/[\gamma(1 - v_x V/c^2)]$, which depends on v_x; since the two balls have different v_x, their y-velocities are changed by different factors. One then finds that the total y-component of $m\mathbf{v}$ is *not* conserved in S'. If we kept $\mathbf{p} = m\mathbf{v}$, momentum conservation would hold in one frame only – violating Postulate 1.

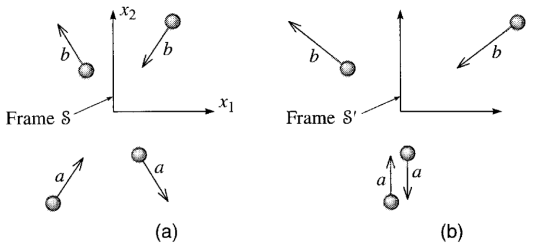


Figure 27: An elastic collision of equal masses seen (a) in frame S and (b) in a frame S' moving along x_1. With the Newtonian definition $\mathbf{p} = m\mathbf{v}$, momentum conserved in S is not conserved in S'.

Source: J. R. Taylor, Classical Mechanics*, Fig. 15.11*

The cure is to use *proper time* τ (the same for all observers) in place of t when defining velocity.

23 Four-velocity, four-momentum and relativistic momentum

Mass, four-velocity, four-momentum.

- **Mass** m: the *rest mass*, an invariant (same in all frames). We do not use a "relativistic mass".
- **Four-velocity**: $u^\mu = \dfrac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \gamma\,(c,\ \mathbf{v})$, since $\mathrm{d}\tau = \mathrm{d}t/\gamma$. It is a four-vector, and $u \cdot u = \gamma^2(v^2 - c^2) = -c^2$.
- **Four-momentum**: $p^\mu = m\,u^\mu = \big(\gamma mc,\ \gamma m\mathbf{v}\big) = \big(E/c,\ \mathbf{p}\big)$.
- **Relativistic (three-)momentum**: $\mathbf{p} = \gamma m\mathbf{v} = \dfrac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}}$.

Law: in an isolated system the total four-momentum is conserved. Because p^μ is a four-vector, if it is conserved in one frame it is conserved in all.

For $v \ll c$, $\gamma \approx 1$ and $\mathbf{p} \approx m\mathbf{v}$, so Newton is recovered. As $v \to c$, $p \to \infty$: an infinite momentum (and energy) would be needed to reach c.

24 Energy and mass–energy equivalence

The conserved time component of p^μ is $p^0 = \gamma mc$. Taylor (§15.13) identifies cp^0 as the energy. Expanding for small v:

$$cp^0 = \gamma mc^2 = mc^2\Big(1 + \frac{v^2}{2c^2} + \frac{3v^4}{8c^4} + \cdots\Big) = mc^2 + \tfrac{1}{2}mv^2 + \cdots$$

The second term is the familiar kinetic energy, so γmc^2 is the relativistic *total energy*.

Relativistic energy.

$$E = \gamma mc^2, \qquad E_0 = mc^2 \text{ (rest energy)}, \qquad T = E - mc^2 = (\gamma - 1)mc^2 \text{ (kinetic energy)}.$$

$$\boxed{E^2 = (pc)^2 + (mc^2)^2}, \qquad \mathbf{v} = \frac{\mathbf{p}\,c^2}{E}, \qquad \beta = \frac{pc}{E}, \qquad \gamma = \frac{E}{mc^2}.$$

E, pc and mc^2 form a right-angled triangle with E as hypotenuse (Fig. 28).

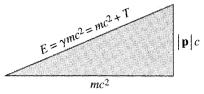


Figure 28: E, mc^2 and $|\mathbf{p}|c$ are related like the sides of a right triangle with E as the hypotenuse. *Source: J. R. Taylor,* Classical Mechanics*, Fig. 15.12*

Derivation [5 marks]: $E^2 = (pc)^2 + (mc^2)^2$ from the four-momentum

$p \cdot p = -(p^0)^2 + \mathbf{p}^2 = -\dfrac{E^2}{c^2} + p^2$. Evaluate the same invariant in the particle's rest frame, where $\mathbf{p} = 0$ and $E = mc^2$: $p \cdot p = -m^2c^2$. Since $p \cdot p$ is invariant,

$$-\frac{E^2}{c^2} + p^2 = -m^2c^2 \quad \Rightarrow \quad E^2 = p^2c^2 + m^2c^4. \qquad \blacksquare$$

(Directly: $E^2 - p^2c^2 = \gamma^2 m^2 c^4(1 - v^2/c^2) = m^2c^4$.)

24.1 Mass–energy equivalence

Rest energy mc^2 is a real, convertible form of energy. Energy conservation now reads $\sum E = \sum(mc^2 + T) = \text{const}$, and rest mass need *not* be conserved:

$$\Delta M\, c^2 = -\Delta T \qquad \text{(Taylor Eq. 15.79)}.$$

- Any energy added to a body at rest (heat, internal excitation) increases its mass by $\Delta m = \Delta E/c^2$.
- In nuclear reactions and particle decays, a decrease of rest mass appears as kinetic energy (and vice versa in collisions that create particles).

- In chemistry the mass changes are only $\sim 10^{-10}$ of the masses – far too small to weigh, which is why mass conservation was believed exact for centuries.

Solved Example 23: Mass change in an atom (Taylor Ex. 15.7)

In the Franck–Hertz experiment a mercury atom (mass $\approx 200\,\mathrm{u}$) absorbs $4.9\,\mathrm{eV}$ from an electron. Its mass increases by $\Delta m = 4.9\,\mathrm{eV}/c^2 = 8.7\times10^{-36}\,\mathrm{kg}$, a fraction $4.9/(200\times931.5\times10^6) = 2.6\times10^{-11}$ of its mass.

Solved Example 24: Binding energy of the deuteron

$m_p = 1.007276\,\mathrm{u}$, $m_n = 1.008665\,\mathrm{u}$, $m_d = 2.013553\,\mathrm{u}$, and $1\,\mathrm{u}\,c^2 = 931.5\,\mathrm{MeV}$. Mass defect $\Delta m = m_p + m_n - m_d = 0.002388\,\mathrm{u}$, so binding energy $= 0.002388\times931.5 = 2.22\,\mathrm{MeV}$. A γ-ray of at least this energy is needed to break a deuteron apart.

Solved Example 25: The Sun loses mass

The solar luminosity is $L = 3.83\times10^{26}\,\mathrm{W}$. Rate of loss of rest mass: $\mathrm{d}m/\mathrm{d}t = L/c^2 = 3.83\times10^{26}/(9\times10^{16}) = 4.3\times10^9\,\mathrm{kg\,s^{-1}}$ – over four million tonnes every second, converted to sunlight by hydrogen fusion.

24.2 Units used in particle physics

Energies in eV, keV, MeV, GeV ($1\,\mathrm{eV} = 1.602\times10^{-19}\,\mathrm{J}$); momenta in MeV/c; masses in MeV/c^2. Electron: $m_e c^2 = 0.511\,\mathrm{MeV}$; proton: $m_p c^2 = 938.3\,\mathrm{MeV}$. In these units $E^2 = (pc)^2 + (mc^2)^2$ is used directly with numbers in MeV.

Solved Example 26: An electron accelerated through 1 MV

$T = 1.000\,\mathrm{MeV}$, so $E = T + m_e c^2 = 1.511\,\mathrm{MeV}$.

$$\gamma = \frac{E}{mc^2} = \frac{1.511}{0.511} = 2.96, \quad \beta = \sqrt{1-1/\gamma^2} = 0.941, \quad pc = \sqrt{1.511^2 - 0.511^2} = 1.42\,\mathrm{MeV},\ p = 1.42\,\mathrm{MeV}/c.$$

Classical $\tfrac12 mv^2 = 1\,\mathrm{MeV}$ would give $v = 2c$ – impossible.

Solved Example 27: Energy and momentum of an electron (Taylor Ex. 15.8)

An electron has $T = 0.8\,\mathrm{MeV}$. Then $E = 1.3\,\mathrm{MeV}$, $pc = \sqrt{1.3^2 - 0.5^2} = 1.2\,\mathrm{MeV}$, and $\beta = pc/E = 0.92$.

25 Force in relativity

The most useful definition (Taylor §15.15) keeps Newton's form with the new momentum:

Relativistic force.

$$\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\big(\gamma m\mathbf{v}\big), \qquad \frac{\mathrm{d}E}{\mathrm{d}t} = \mathbf{F}\cdot\mathbf{v} \quad \text{(work–energy theorem: } \mathrm{d}T = \mathbf{F}\cdot\mathrm{d}\mathbf{r}\text{)}.$$

With this definition the Lorentz force $\mathbf{F} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B})$ is exactly right at all speeds.

Derivation [5 marks]: Work–energy theorem

Differentiate $E^2 = p^2c^2 + m^2c^4$ with respect to t (m constant): $2E\,\mathrm{d}E/\mathrm{d}t = 2c^2\,\mathbf{p}\cdot\mathrm{d}\mathbf{p}/\mathrm{d}t = 2c^2\,\mathbf{p}\cdot\mathbf{F}$. Using $\mathbf{p}c^2/E = \mathbf{v}$: $\mathrm{d}E/\mathrm{d}t = \mathbf{v}\cdot\mathbf{F}$. Multiplying by $\mathrm{d}t$, $\mathrm{d}E = \mathbf{F}\cdot\mathrm{d}\mathbf{r}$; since $E = mc^2 + T$, $\mathrm{d}T = \mathbf{F}\cdot\mathrm{d}\mathbf{r}$. ■

Force and acceleration are not parallel. Differentiating $\mathbf{p} = \gamma m\mathbf{v}$:

$$\mathbf{F} = \gamma m\mathbf{a} + \gamma^3 m\frac{(\mathbf{v}\cdot\mathbf{a})}{c^2}\mathbf{v}.$$

- Force perpendicular to $\mathbf{v}$ (e.g. magnetic force): $F = \gamma ma$.
- Force parallel to $\mathbf{v}$: $F = \gamma^3 ma$.

A fast particle is much harder to accelerate along its motion than sideways – and can never reach c.

Solved Example 28: Motion under a constant force (Taylor Ex. 15.11)
A particle starts from rest under a constant force F. Then $p = Ft$, and from $p = \gamma mv$,

$$v = \frac{Ft/m}{\sqrt{1+(Ft/mc)^2}} \longrightarrow c \text{ as } t\to\infty.$$

For an electron in a field of $10^6\,\mathrm{V/m}$, $F = 1.6\times10^{-13}\,\mathrm{N}$ and $mc = 2.73\times10^{-22}\,\mathrm{kg\,m/s}$. At $t = mc/F = 1.7\,\mathrm{ns}$ Newton would predict $v = c$; relativity gives $v = c/\sqrt{2} = 0.71c$.

Four-force. $K^\mu = \mathrm{d}p^\mu/\mathrm{d}\tau = \gamma\big(\mathbf{F}\cdot\mathbf{v}/c,\ \mathbf{F}\big)$ is a four-vector (Taylor Eq. 15.108), so forces can be transformed between frames with Λ. A useful special case: for a particle momentarily at rest in S', the force component along the relative motion is the same in both frames, $F_\parallel = F'_\parallel$, while perpendicular components are reduced, $F_\perp = F'_\perp/\gamma$ (used in Unit 2).

26 Transformation of momentum and energy

Since $p^\mu = (E/c, \mathbf{p})$ is a four-vector, it transforms exactly like $(ct, \mathbf{r})$:

Lorentz transformation of momentum and energy.

$$p'_x = \gamma\Big(p_x - \frac{VE}{c^2}\Big), \qquad p'_y = p_y, \qquad p'_z = p_z, \qquad E' = \gamma\,(E - Vp_x).$$

Invariant: $E^2 - p^2c^2 = E'^2 - p'^2c^2 = (mc^2)^2$. For a system of particles the same holds for the totals; $\sqrt{(\sum E)^2 - (\sum \mathbf{p})^2c^2}/c^2$ is the **invariant mass** of the system.

Solved Example 29: Transforming a proton's energy and momentum
A proton moves along x with $T = 1.0\,\mathrm{GeV}$ in S: $E = 1.938\,\mathrm{GeV}$, $pc = \sqrt{1.938^2 - 0.938^2} = 1.696\,\mathrm{GeV}$. Frame S' moves along x at $\beta = 0.6$ ($\gamma = 1.25$):

$$E' = 1.25\,(1.938 - 0.6\times1.696) = 1.151\ \mathrm{GeV}, \qquad p'c = 1.25\,(1.696 - 0.6\times1.938) = 0.667\ \mathrm{GeV}.$$

Check: $E'^2 - (p'c)^2 = 1.325 - 0.444 = 0.880 = (0.938)^2$ ✓.

27 Massless particles: the photon

From $E^2 = (pc)^2 + (mc^2)^2$ and $v = pc^2/E$: if $m = 0$ then $E = pc$ and $v = c$. Conversely, anything moving at c must be massless. The photon is such a particle (Taylor §15.16):

$$E = pc = hf, \qquad p = \frac{h}{\lambda}, \qquad p^\mu = \Big(\frac{E}{c},\ \mathbf{p}\Big), \quad p\cdot p = 0.$$

A photon has no rest frame (it is never "at rest").

Solved Example 30: Photons

(a) A 500 nm photon: $E = hc/\lambda = 2.48\,\text{eV}$, $p = h/\lambda = 1.33 \times 10^{-27}\,\text{kg m/s}$.

(b) A neutral pion ($m_\pi c^2 = 135\,\text{MeV}$) at rest decays into two photons. Momentum conservation requires equal and opposite photons, so each has $E = 67.5\,\text{MeV}$.

Compton scattering (result only): conservation of four-momentum for a photon scattering off a free electron gives $\lambda' - \lambda = \dfrac{h}{m_e c}(1 - \cos\theta)$, with $h/m_e c = 2.43\,\text{pm}$ – one of the best confirmations of relativistic dynamics for photons.

28 Collisions

In any collision total energy and total momentum are conserved; kinetic energy and total rest mass need not be.

Solved Example 31: Two lumps of putty (Taylor Ex. 15.9)

Two lumps, each of mass m, approach head-on, each at $0.6c$, and stick together. By symmetry the final lump is at rest, so all energy becomes rest energy:

$$Mc^2 = 2\gamma mc^2 = 2(1.25)mc^2 \;\Rightarrow\; M = 2.5m.$$

The final mass exceeds $2m$: the lost kinetic energy $2(\gamma - 1)mc^2 = 0.5mc^2$ has become rest energy (internal heat).

Solved Example 32: Threshold for pair production

A photon can create an electron–positron pair only if enough energy is available in the centre-of-mass frame. Near a heavy nucleus (which absorbs momentum but little energy) the threshold is $E_\gamma \geq 2m_e c^2 = 1.022\,\text{MeV}$. In empty space a single photon can *never* create a pair: the invariant mass of the photon is zero ($p \cdot p = 0$), whereas the invariant mass of an electron–positron pair is at least $2m_e$. Since total four-momentum is conserved, the invariant mass before and after must be equal – impossible.

Summary of relativistic dynamics.

Momentum	$\mathbf{p} = \gamma m \mathbf{v}$
Energy	$E = \gamma mc^2 = mc^2 + T$
Energy–momentum	$E^2 = (pc)^2 + (mc^2)^2, \quad \mathbf{v} = \mathbf{p}c^2/E$
Four-momentum	$p^\mu = (E/c, \mathbf{p}), \quad p \cdot p = -m^2c^2$
Force	$\mathbf{F} = \mathrm{d}\mathbf{p}/\mathrm{d}t, \quad \mathrm{d}E/\mathrm{d}t = \mathbf{F} \cdot \mathbf{v}$
Transformation	$p'_x = \gamma(p_x - VE/c^2), \;\; E' = \gamma(E - Vp_x)$
Photon	$E = pc = hf$

Part D Gravity and Space-Time [Unit 3]

Both sections of Part D belong to Unit 3. Special relativity treats only inertial frames and leaves gravity out. Einstein's next question (1907) was: what happens in accelerated frames and in gravity? The answer begins with a single, simple principle.

29 The principle of equivalence and general relativity [Unit 3]

29.1 Gravitational and inertial mass

Mass appears in Newtonian physics in two different roles: *inertial mass* m_i in $F = m_i a$ and *gravitational mass* m_g in $F = GMm_g/r^2$. Experiment (Galileo, Newton, Eötvös 1889, and modern tests to 1 part in 10^{15}) shows $m_i = m_g$, which is why all bodies fall with the same acceleration. Newton regarded this as a coincidence; Einstein took it as a clue.

29.2 Einstein's elevator

In 1907 Einstein had what he called "the happiest thought of my life": a person falling freely does not feel his own weight.

- A closed laboratory resting on the Earth (gravity g downwards) and an identical laboratory in deep space accelerating upwards at g: dropped balls, pendulums, scales – every experiment gives the same result (Fig. 29).
- A laboratory falling freely in a uniform gravitational field and a laboratory floating in deep space: in both, objects float and move in straight lines (Fig. 30).

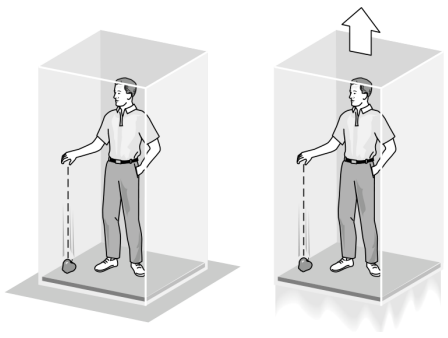


Figure 29: Gravity is equivalent to an accelerating laboratory: (a) a laboratory on Earth, (b) a laboratory accelerating at g in space.
Source: Carroll & Ostlie, Ch. 17, Fig. 6

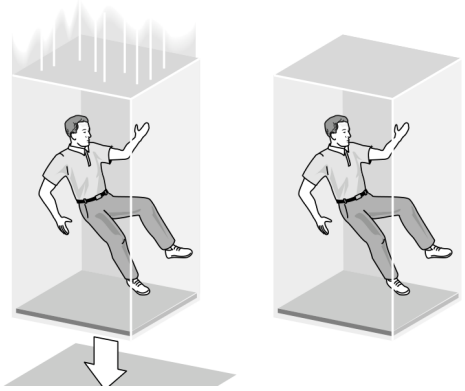


Figure 30: Gravity abolished: (a) a laboratory in free fall, (b) a laboratory floating in space.
Source: Carroll & Ostlie, Ch. 17, Fig. 7

Principle of equivalence.

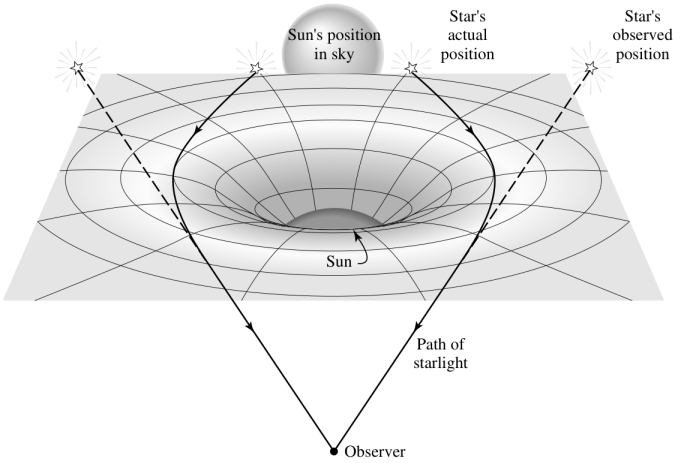
Weak form: $m_i = m_g$; all bodies fall the same way in a gravitational field.

Einstein's (strong) form: In a small enough region of space-time, a uniform gravitational field is completely equivalent to a uniformly accelerated frame of reference. No experiment of *any* kind – mechanical, optical, electromagnetic – performed inside a closed laboratory can

distinguish the two. Equivalently: in a freely falling laboratory the laws of special relativity hold (locally); freely falling frames are the true *local inertial frames*.

29.3 Consequences

(1) Light bends in gravity. In an accelerating rocket a horizontal light beam curves downwards relative to the walls (the floor rises while the light crosses). By equivalence, light must also bend in a gravitational field. General relativity predicts a deflection of $1.75''$ for starlight grazing the Sun. Eddington's eclipse expeditions of 1919 confirmed it (Figs. 31).



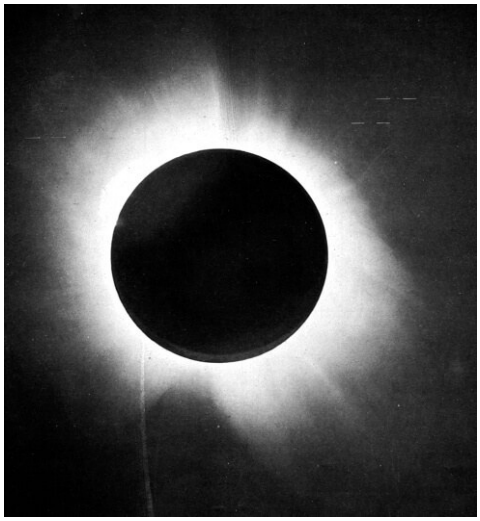


Figure 31: **Left:** starlight passing the Sun follows the curvature of space-time, so the star is seen displaced away from the Sun.
Source: Carroll & Ostlie, Ch. 17, Fig. 5 **Right:** photograph of the total solar eclipse of 29 May 1919 used by Eddington and Dyson to measure the deflection.
Source: Wikimedia Commons, "1919 eclipse positive.jpg" (public domain; F. W. Dyson, A. S. Eddington & C. Davidson, 1920)

(2) Clocks run at different rates at different heights – the gravitational red shift (next section).

(3) Gravity is geometry. Since all bodies fall the same way regardless of their mass or composition, Einstein interpreted gravity not as a force but as the *curvature of space-time*. Freely falling bodies and light follow the "straightest possible" paths (geodesics) in curved space-time. In J. A. Wheeler's words: *matter tells space-time how to curve, and space-time tells matter how to move.* The field equations of general relativity (1915) relate the curvature of space-time to the energy and momentum of matter; in weak fields they reduce to Newton's gravity.

Classic tests of general relativity.

Perihelion of Mercury	Extra advance of $43''$ per century, unexplained by Newton, predicted exactly (1915).
Bending of light	$1.75''$ at the Sun's edge; 1919 eclipse; today gravitational lenses (Unit 3, Part B).
Gravitational red shift	Pound–Rebka (1960); white dwarfs; GPS satellites.
Radar echo delay	Shapiro (1964 onwards), signals passing the Sun delayed.
Gravitational waves	Binary pulsar (1974); direct detection by LIGO (2015).

30 Gravitational red shift [Unit 3]

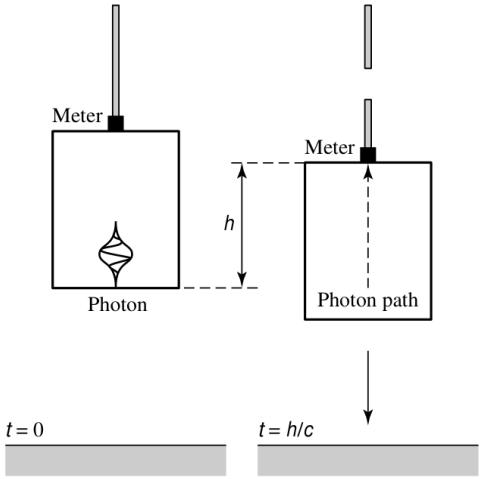


Figure 32: A photon emitted from the floor of a laboratory accelerating upwards at g reaches the ceiling (height h) after time h/c, by which time the ceiling has acquired an extra velocity gh/c away from the emission point.
Source: B. W. Carroll & D. A. Ostlie, An Introduction to Modern Astrophysics*, Ch. 17, Fig. 11*

Derivation [5 marks]: Gravitational red shift from the equivalence principle

(i) Replace the uniform field g by a rocket accelerating upwards at g in space. A photon of frequency ν_0 leaves the floor at $t = 0$, when the rocket is (momentarily) at rest in an inertial frame.

(ii) It reaches the ceiling, height h, after $t \approx h/c$. By then the ceiling moves upward with speed $\Delta v = gt = gh/c$, i.e. *away* from the point of emission.

(iii) The receiver therefore sees a Doppler red shift (first order, $\Delta v \ll c$):

$$\nu = \nu_0\Big(1 - \frac{\Delta v}{c}\Big) = \nu_0\Big(1 - \frac{gh}{c^2}\Big), \qquad \frac{\Delta\nu}{\nu_0} = -\frac{gh}{c^2}.$$

(iv) By the principle of equivalence, the same shift must occur for light climbing a height h in a gravitational field g. Since $gh = \Delta\Phi$ (the difference in gravitational potential),

$$\boxed{\frac{\Delta\nu}{\nu} = -\frac{\Delta\Phi}{c^2}} \qquad \blacksquare$$

Energy check: a photon of energy $h\nu$ has an equivalent mass $h\nu/c^2$; climbing a height h it loses energy $(h\nu/c^2)gh$, so $\Delta\nu/\nu = -gh/c^2$, the same answer.

Gravitational red shift.
Light climbing out of a gravitational potential well is shifted to lower frequency. From the surface of a body of mass M and radius R to a distant observer (weak field):

$$z = \frac{\Delta\lambda}{\lambda} \approx \frac{GM}{Rc^2}.$$

Exact (Schwarzschild) result: $1+z=\left(1-\dfrac{2GM}{Rc^2}\right)^{-1/2}$.

Equivalently, **clocks lower in a gravitational potential run slower**: $\Delta\tau_{\rm low}/\Delta\tau_{\rm high}=1-\Delta\Phi/c^2$. Unlike special-relativistic time dilation, this effect is *not* mutual: both observers agree which clock is slower.

30.1 Experimental confirmation

Pound–Rebka experiment (1960). Gamma rays of 14.4 keV from ^{57}Fe were sent up (and down) the 22.5 m tower of the Jefferson Laboratory at Harvard. The expected fractional shift is only $gh/c^2=2.5\times10^{-15}$, but the extremely sharp Mössbauer resonance made it measurable: the source was moved slowly to compensate the shift by a Doppler shift. Result: agreement with theory to 10% (1% by Pound and Snider, 1964).

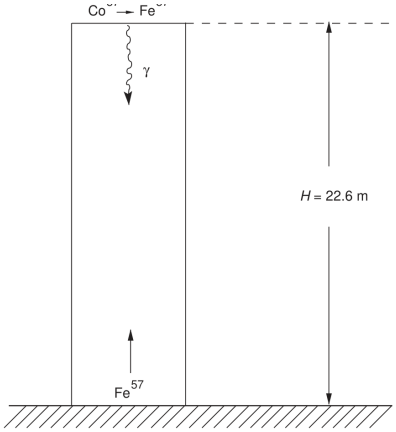


Figure 33: Schematic of the Pound–Rebka experiment: γ-rays from ^{57}Co$\rightarrow$ ^{57}Fe fall (or rise) through a height $H\approx22.6$ m and are detected by resonant absorption in ^{57}Fe.
Source: J. V. Narlikar, An Introduction to Relativity*, Fig. 10.3*

White dwarfs. Spectral lines from Sirius B and 40 Eridani B are red-shifted by $z\sim10^{-4}$, as predicted for their large M/R.

GPS. Satellite clocks run fast relative to ground clocks because they are higher in the Earth's potential (general relativity), and slow because they move (special relativity).

Solved Example 33: The Pound–Rebka shift

$$\frac{\Delta\nu}{\nu}=\frac{gh}{c^2}=\frac{(9.8)(22.5)}{(3\times10^8)^2}=2.45\times10^{-15}.$$

Solved Example 34: Red shift from the Sun and from a white dwarf

Sun: $\dfrac{GM_\odot}{R_\odot c^2}=\dfrac{(6.67\times10^{-11})(1.99\times10^{30})}{(6.96\times10^8)(9\times10^{16})}=2.1\times10^{-6}$. A 500 nm line is shifted by only 0.001 nm.

White dwarf ($M=M_\odot$, $R=5.8\times10^6$ m): $z=1.33\times10^{20}/(5.8\times10^6\times9\times10^{16})=2.5\times10^{-4}$, equivalent to a Doppler velocity $cz\approx76\,\mathrm{km\,s^{-1}}$.

Neutron star ($M=1.4M_\odot$, $R=10$ km): $2GM/Rc^2=0.41$, so the weak-field formula fails; the exact formula gives $1+z=(1-0.41)^{-1/2}=1.31$, i.e. $z\approx0.31$.

Solved Example 35: GPS clocks

GPS satellites orbit at $r=2.66\times10^7$ m with speed $3.87\,\mathrm{km\,s^{-1}}$; Earth's radius is $R=6.37\times10^6$ m and $GM_E=3.99\times10^{14}\,\mathrm{m^3\,s^{-2}}$.

Gravitational (GR): $\dfrac{\Delta\Phi}{c^2} = \dfrac{GM_E}{c^2}\left(\dfrac{1}{R} - \dfrac{1}{r}\right) = 5.3 \times 10^{-10}$ $\Rightarrow$ satellite clock gains $5.3 \times 10^{-10} \times 86400\,\mathrm{s} = 45.7\,\mu\mathrm{s/day}$.

Velocity (SR): $\frac{1}{2}v^2/c^2 = 8.3 \times 10^{-11}$ $\Rightarrow$ loses $7.2\,\mu\mathrm{s/day}$.

Net: the satellite clock runs **fast by about 38 μs per day**. Light travels 11 km in 38 μs, so uncorrected GPS positions would drift by ~ 10 km every day.

Unit 3 (Part A) – summary of the geometric picture.

- Space-time diagram: events are points, histories are world lines, light moves at 45°.
- A moving frame's ct' and x' axes tilt symmetrically towards the light line: this is the relativity of simultaneity, and it produces time dilation and length contraction.
- Interval $s^2 = \Delta r^2 - c^2\Delta t^2$ is invariant; its sign gives timelike ($s^2 < 0$, absolute order, causal), lightlike ($s^2 = 0$) or spacelike ($s^2 > 0$, order frame-dependent, no causal link).
- Four-vectors and tensors transform with Λ; laws written as tensor equations hold in all frames.
- Twin paradox: proper time $\int \mathrm{d}t/\gamma$ is greatest along the inertial (straight) world line.
- Equivalence principle: gravity $\equiv$ acceleration locally; gravity is the curvature of space-time.
- Gravitational red shift: $\Delta\nu/\nu = -\Delta\Phi/c^2 \approx -gh/c^2$; $z \approx GM/Rc^2$.